The **Evaluation** phase focused on assessing the accuracy, interpretability, and practical usefulness of the generated clusters. Cluster validation metrics, including the Calinski–Harabasz Index, Davies–Bouldin Index, and Silhouette Score, were used to measure clustering quality. In addition, academic staff, guidance counselors, and IT personnel reviewed the clustering results to determine whether the identified patterns accurately reflected student performance and were suitable for academic decision support.

Finally, in the **Deployment** phase, the validated clustering model was integrated into a web-based system with a user-friendly dashboard. The system allowed authorized users to upload ICAT scores, generate and visualize clusters, and access individual student profiles. This phase transformed the clustering model into a practical tool for real-time academic analysis and data-driven planning at ISPSC Tagudin Campus.

Guided by the CRISP-DM framework, the **Business Understanding** phase focused on identifying meaningful groupings from ICAT scores that could support personalized academic guidance. This phase involved defining project objectives such as determining student eligibility for board courses and identifying cognitive strengths and weaknesses. Establishing these objectives ensured that the system aligned with institutional priorities, including curriculum improvement and the effective use of ICAT data for informed academic decision-making.

The **Data Understanding** phase involved the collection of ICAT results for the academic years 2024–2025 and 2025–2026 from the ISPSC Guidance Office. During this phase, the dataset was examined to identify issues such as missing values, inconsistencies, and outliers. Gaining a comprehensive understanding of the dataset at this stage ensured the reliability and validity of subsequent analyses.

In the **Data Preparation** phase, the dataset was cleaned, normalized, and transformed into an analyzable format. This process included removing incomplete records, selecting the six ICAT cognitive components, and normalizing the scores. Proper data preparation was essential to produce consistent and reliable clustering results.

The **Modeling** phase applied the K-Means clustering algorithm to the prepared dataset to identify natural groupings among students. The Elbow Method was used to determine the optimal number of clusters by identifying the point at which additional clusters no longer significantly improved the model's performance. Based on the clustering output, learner groupings such as All-Around Strong, Cognitive-Oriented, and Practical-Oriented students emerged.

developing a web-based dashboard for data uploading, analysis, and visualization. The adoption of this model addressed the institution's need for improved data utilization and evidence-based academic decision-making while ensuring data accuracy, integrity, and ethical handling of student information, as illustrated in Figure 2.

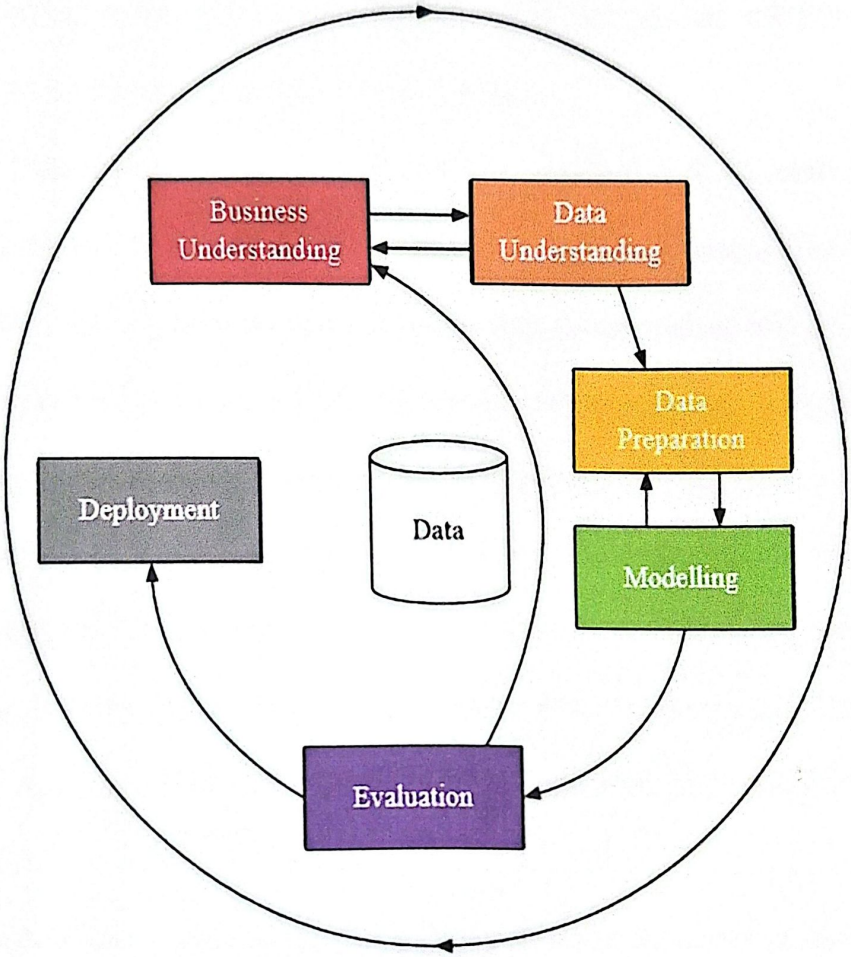


Figure 2. CRISP-DM (Cross-Industry Standard Process for Data Mining)

Figure 2 presents the CRISP-DM model adopted in this study. The model consists of six iterative phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. These phases guided the entire development process of the clustering system from conceptualization to implementation.

BIOGRAPHICAL SKETCH

Azriel Jon M. Yukee, a 22-year-old student of Ilocos Sur Polytechnic State College (ISPSC) – Tagudin Campus, is a young individual driven by passion and determination in the field of Information Technology. He is the proud son of Mr. Froilan M. Yukee and Mrs. Yves Jill M. Yukee, who have continuously supported and guided him throughout his journey. He began his academic path with excellence at the Regional Science High School for Region I, where he was recognized as one of the most outstanding students. His dedication to academics and eagerness to learn provided him with a strong foundation in science and technology, shaping his confidence to pursue IT as his chosen field. From an early age, Azriel has shown curiosity and determination in learning new skills, particularly in programming and technology.

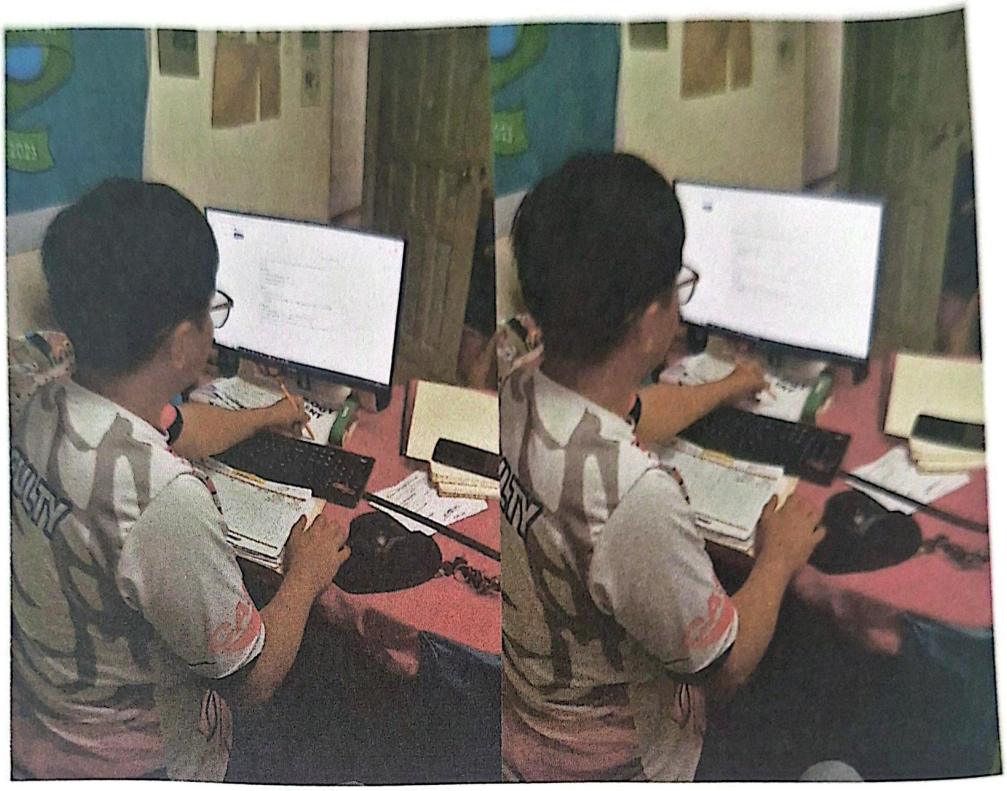
He dreams of excelling as an IT professional, open to any specialty where his dedication, persistence, and hard work can truly make a difference. Azriel is not only eager to learn but also embraces challenges as opportunities for growth. He believes that success comes through perseverance, humility, and a constant desire for improvement. His natural leadership abilities and strong social skills allow him to work effectively with others, making him both a collaborative team player and a capable leader. In every endeavor, Azriel strives to balance technical excellence with genuine human connection. His journey reflects the values of resilience, curiosity, and service.

A.J.M.Y

APPENDIX O

DOCUMENTATION

On **August 8, 2025**, we asked Sir Sonny how the ICAT scores are interpreted in percentiles. He mentioned that a score in the 33rd percentile represents the student's relative performance on the test.



On **October 13, 2025**, the project manager presented the system to the Guidance Counselor, providing a concise overview of its purpose and the specific features designed to support student monitoring and record management. The project manager demonstrated how to navigate the dashboard, access student information, generate reports, and utilize tools for tracking behavior or academic performance. Throughout the session, they explained each function clearly, addressed the Guidance Counselor's questions, and ensured the Guidance Counselor felt confident and prepared to use the system effectively.

APPENDIX N

CERTIFICATE OF STATISTICIAN

ILOCOS SUR POLYTECHNIC STATE COLLEGE

TAGUDIN CAMPUS

Tagudin, Ilocos Sur

COLLEGE OF ARTS AND SCIENCES

CERTIFICATION

This is to certify that I have checked the capstone project manuscript copy of **YUKEE, AZRIEL JON M., BONIFACIO, CHRISTINE LORENCE., SALVADOR, JOHN MARK A., MACABITAS, ARIES P.** entitled **"CLUSTERING STUDENT ICAT SCORE USING MACHINE LEARNING ALGORITHM"** which has been required for graduation in the BSIT program.

This certification is issued on this 22nd day of December 2025, at ILOCOS SUR POLYTECHNIC STATE COLLEGE TAGUDIN CAMPUS, TAGUDIN, ILOCOS SUR.

GEORGE R. VILLANUEVA JR, DTI

Statistician

student performance trends. This design supports the study's objective of providing a practical, accessible, and data-driven tool for academic guidance and institutional decision-making at ISPSC Tagudin Campus. The overall research process was conducted in iterative phases, including data collection, preprocessing, model development, system implementation, and evaluation, ensuring both technical reliability and contextual relevance.

To evaluate the system's usability, the System Usability Scale (SUS) was utilized as the primary research instrument. The SUS was selected due to its reliability in assessing usability, efficiency, and user satisfaction in system-based applications. The instrument was administered to the Guidance Counselor and Guidance Clerk, who are the intended end users of the system. Despite the limited number of respondents, this approach is methodologically sound, as SUS focuses on usability feedback from actual users, providing context-specific and valid evaluation of the system's effectiveness in meeting the operational needs of the Guidance Office.

Software Model

To ensure that the system development followed a systematic and data-driven approach, this study adopted the Cross-Industry Standard Process for Data Mining (CRISP-DM) as its software development paradigm. CRISP-DM is widely recognized for its structured and iterative framework, which emphasizes problem understanding, data preparation, model development, evaluation, and deployment. Unlike traditional software development models such as Rapid Application Development (RAD) or the Waterfall model, which focus primarily on user interface design or follow a strictly linear sequence of steps, CRISP-DM centers on data analysis and model-driven decision-making. This made it highly suitable for clustering ICAT scores and

APPENDIX M

PLAGIARISM CERTIFICATION

CERTIFICATION ON PLAGIARISM VERIFICATION

This is to certify that the thesis manuscript entitled **"Clustering Student ICAT Score Using Machine Learning Algorithm"** has undergone plagiarism content checking on November 29, 2025, and obtained a similarity index of one point thirty-seven percent (1.37%) using ZeroGPT plagiarism detection software.

Issued this 29th day of NOVEMBER 2025 at Ilocos Sur Polytechnic State College, Tagudin Campus - MIS Office.

JUNEL VENTURA
Person conducted the plagiarism test

APPENDIX L

PLAGIARISM RESULT

ZeroGPT

Your Text is Human Written

1.37%
AI GPT*

Chapter 1

INTRODUCTION

The present scenario of higher education is faced with so much pressure to make decisions about students' aptitudes and cognitive profiles effectively for educational planning and career guidance. Various related studies have established the potential of clustering in educational data analysis. Fathanah (2022) showed that

APPENDIX K

CERTIFICATE OF ENGLISH CRITIC

ILOCOS SUR POLYTECHNIC STATE COLLEGE

TAGUDIN CAMPUS

Tagudin, Ilocos Sur

COLLEGE OF ARTS AND SCIENCES

CERTIFICATION

This is to certify that I have checked the capstone project manuscript copy of **YUKEE, AZRIEL JON M., BONIFACIO, CHRISTINE LORENCE., SALVADOR, JOHN MARK A., MACABITAS, ARIES P.** entitled **"CLUSTERING STUDENT ICAT SCORE USING MACHINE LEARNING ALGORITHM"** which has been required for graduation in the BSIT program.

This certification is issued on this 21st day of November 2025 at Ilocos Sur Polytechnic State College - Tagudin Campus, Tagudin, Ilocos Sur.

RHODA CONEPCION O. VILLAMOR, MSE

English Critic

APPENDIX J

SUS TABULATION

Respondents	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	X(ODD)	X(EVEN)	SUS SCORE
Guidance Associate	2	2	5	5	4	2	5	1	4	2	12	16	70
Guidance Counselor	3	4	4	2	3	7	4	2	4	2	17	17	65

Mean	Grade	Standard Deviation	Minimum	Maximum
67.5	C	3.54	65	70

APPENDIX I

CERTIFICATE OF IMPLEMENTATION

REPUBLIC OF THE PHILIPPINES
ILOCOS SUR POLYTECHNIC STATE COLLEGE
TAGUDIN CAMPUS
TAGUDIN, ILOCOS SUR

CERTIFICATE OF IMPLEMENTATION

This is to certify that the Capstone Project entitled:
"Clustering Student ICAT Score Using Machine Learning Algorithm"
was successfully implemented at Ilocos Sur Polytechnic State College, Tagudin Campus and satisfactorily met the user requirement of the developed system.

Developers:
Azriel Jon M. Yukee
Chistine Lorence H. Bonifacio
John Mark A. Salvador
Aries P. Macabitas

Adviser: Mr. George Villanueva

This certification is issued this 22nd of October, 2025 at Ilocos Sur Polytechnic State College, **Tagudin** Campus, Tagudin, Ilocos Sur.

Dr. Omar C. Valenzuela
Guidance Councelor

be able to use this system.					
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6. I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8. I found this system very awkward to use	☐	☐	☐	☐	☐
9. I felt very confident using this system.	☐	☐	☐	☐	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

APPENDIX H

QUESTIONNAIRE (SUS)

The following questions are designed to evaluate how acceptable and user-friendly the Clustering Student ICAT Score Using Machine Learning Algorithm is. The researchers used the System Usability Scale (SUS) to assess the system's performance and gather feedback for improvement. Please answer honestly based on your experience.

Name (Optional): _____ **Sex:** _____

Position: _____

Rating:

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

Instruction: Your feedback is essential in helping us improve the system and ensure it meets your needs. Please respond honestly based on your experience.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
	1	2	3	4	5
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3. I thought this system was easy to use.	☐	☐	☐	☐	☐
4. I think that I would need the support of a technical person to	☐	☐	☐	☐	☐

APPENDIX G

DEPLOYMENT LETTER

Republic of the Philippines
ILOCOS SUR POLYTECHNIC STATE COLLEGE
College of Arts and Sciences

October 20, 2025

DR. OMAR C. VALENZUELA
Guidance Counselor

Sir:

The undersigned students of the Bachelor of Science in Information Technology (BSIT) program respectfully seek your approval for the deployment of our Capstone Project 2 study entitled "Clustering Student ICAT Score Using Machine Learning Algorithm," to be implemented in the Guidance Office of Ilocos Sur Polytechnic State College (ISPSC) – Tagudin Campus.

The deployment process will include system installation, configuration of the web application, creation of administrative accounts, user orientation, and pilot testing involving selected Guidance Office personnel and faculty members. All data to be processed will be handled with strict confidentiality, following institutional data privacy policies and ethical standards.

We sincerely appreciate your support and approval in deploying our system, which we believe will contribute significantly to ISPSC's ongoing efforts to integrate data-driven solutions in education and improve academic decision-making processes.

Respectfully yours,

AZRIEL JON M. YUKEE
Researcher

CHRISTINE LORENZE H. BONIFACIO
Researcher

JOHN MARK A. SALVADOR
Researcher

ARIES P. MACABITAS
Researcher

Noted:

GEORGE VILLANUEVA JR, DIT
Adviser

JIM-MAR F. DELOS REYES, MIT
Program Head, BSIT

JONALYN S. LACAMBRA, PHD
Dean, College of Arts and Sciences

Approved:

DR. OMAR C. VALENZUELA
Guidance Counselor

VISION

MISSION

CORE VALUES
Innovativeness
Motivation
Professionalism
Excellence
Transparency and Accountability
Unity and Teamwork
Sustainability

MOTTO
Sapientia et Virtus in Prudentia (Wisdom and Virtue for the Future)

DEVELOPMENT GOALS
Onwards UIP

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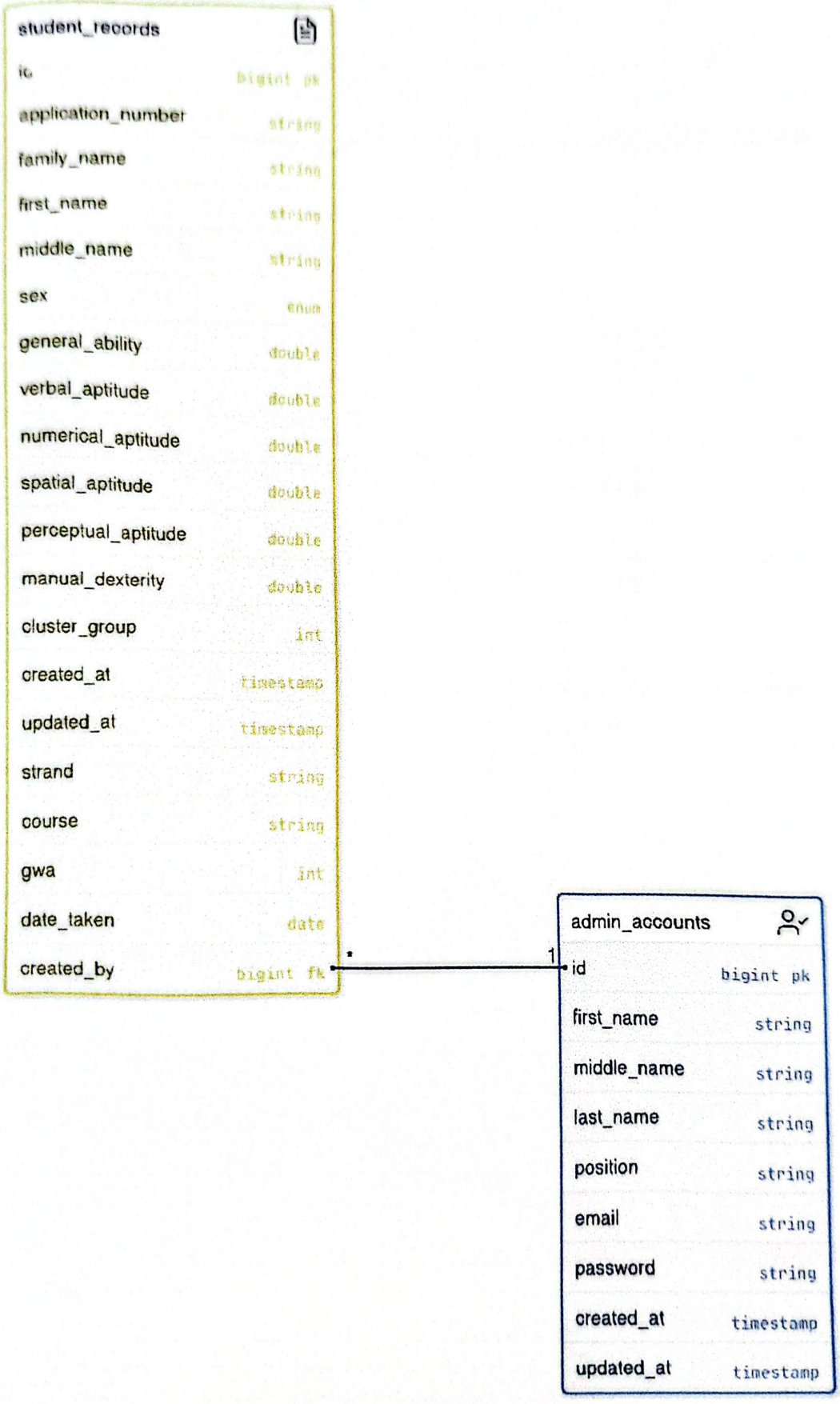
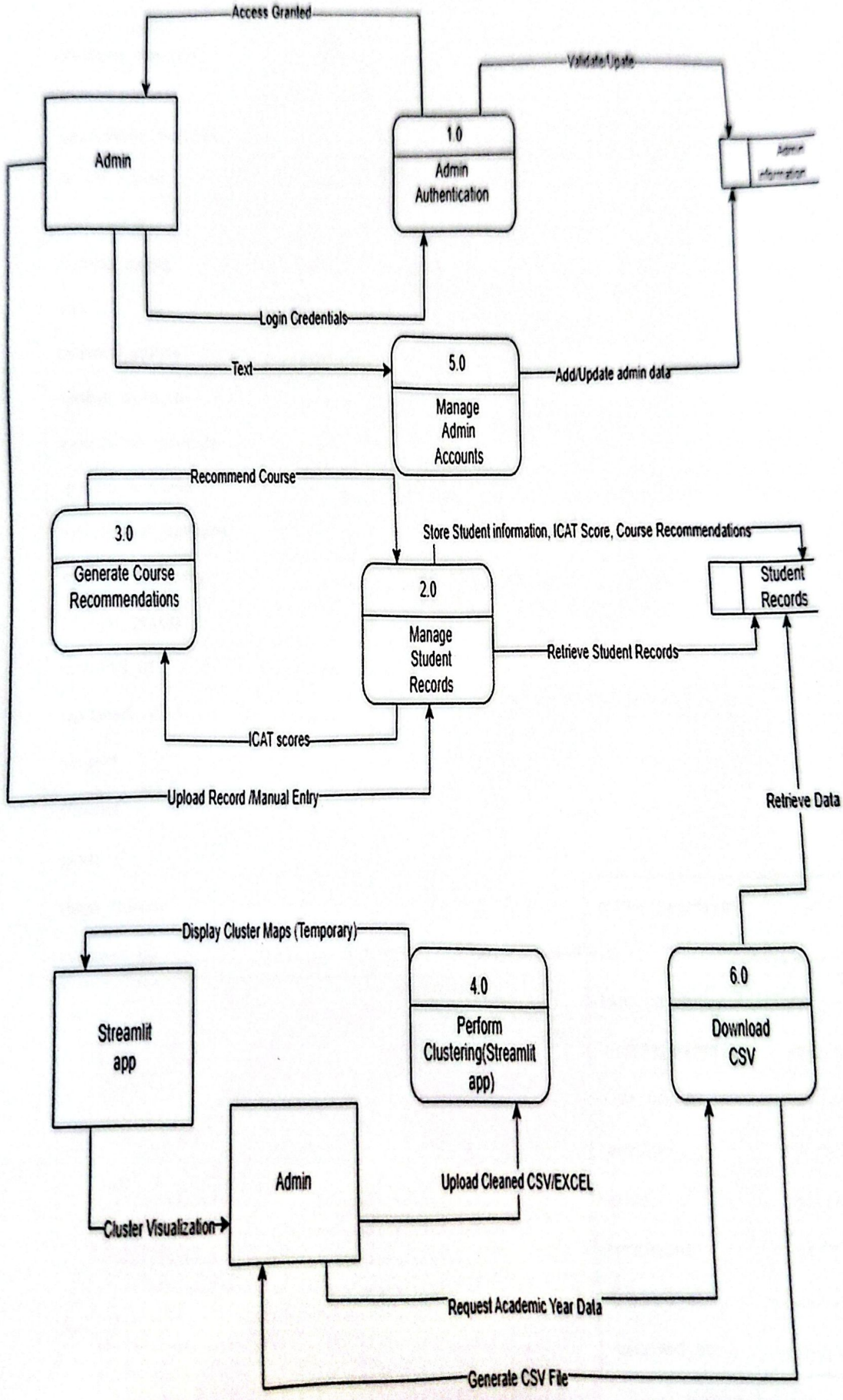
APPENDIX F

ENTITY RELATIONSHIP DIAGRAM



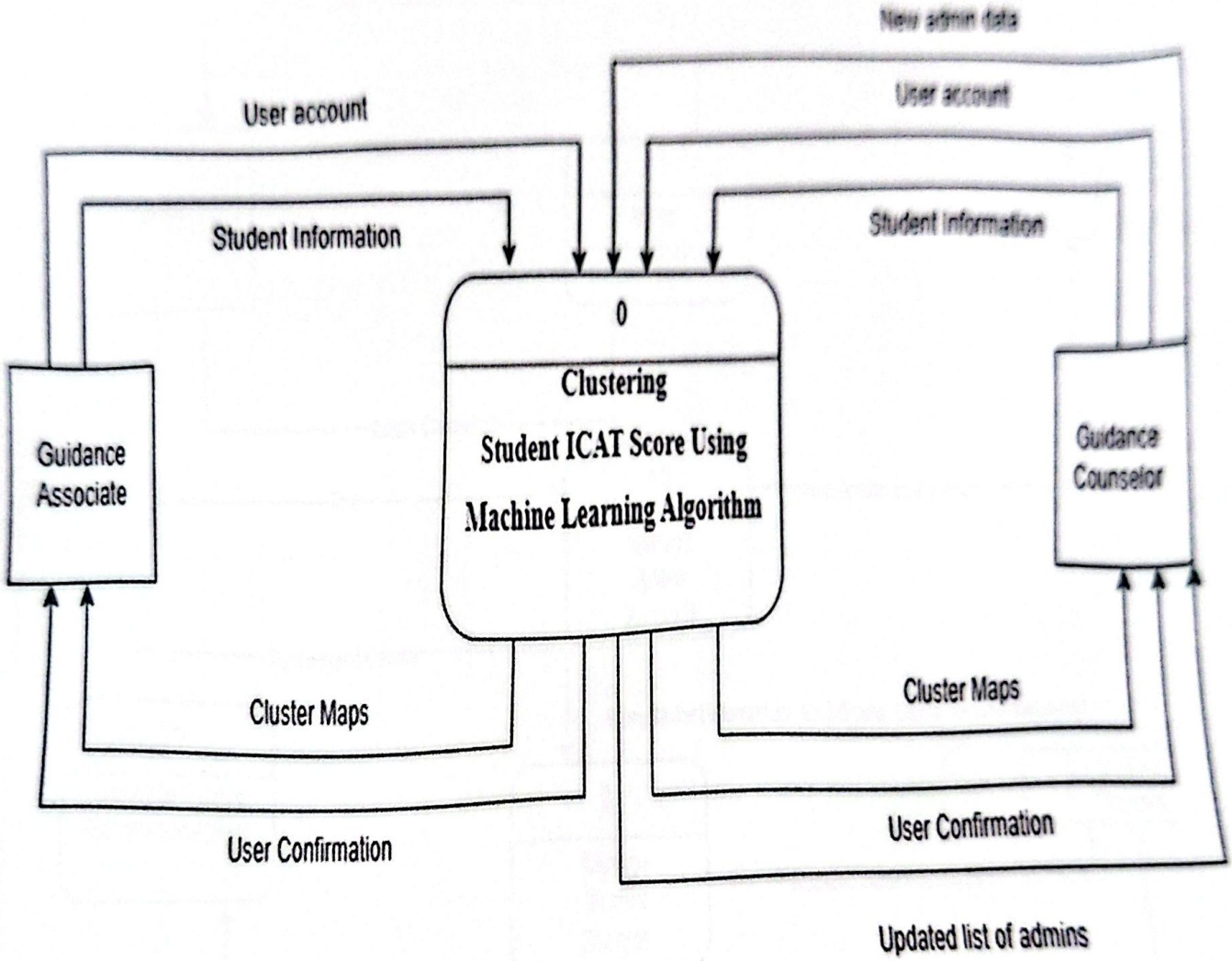
APPENDIX E

DATA FLOW DIAGRAM LEVEL 1



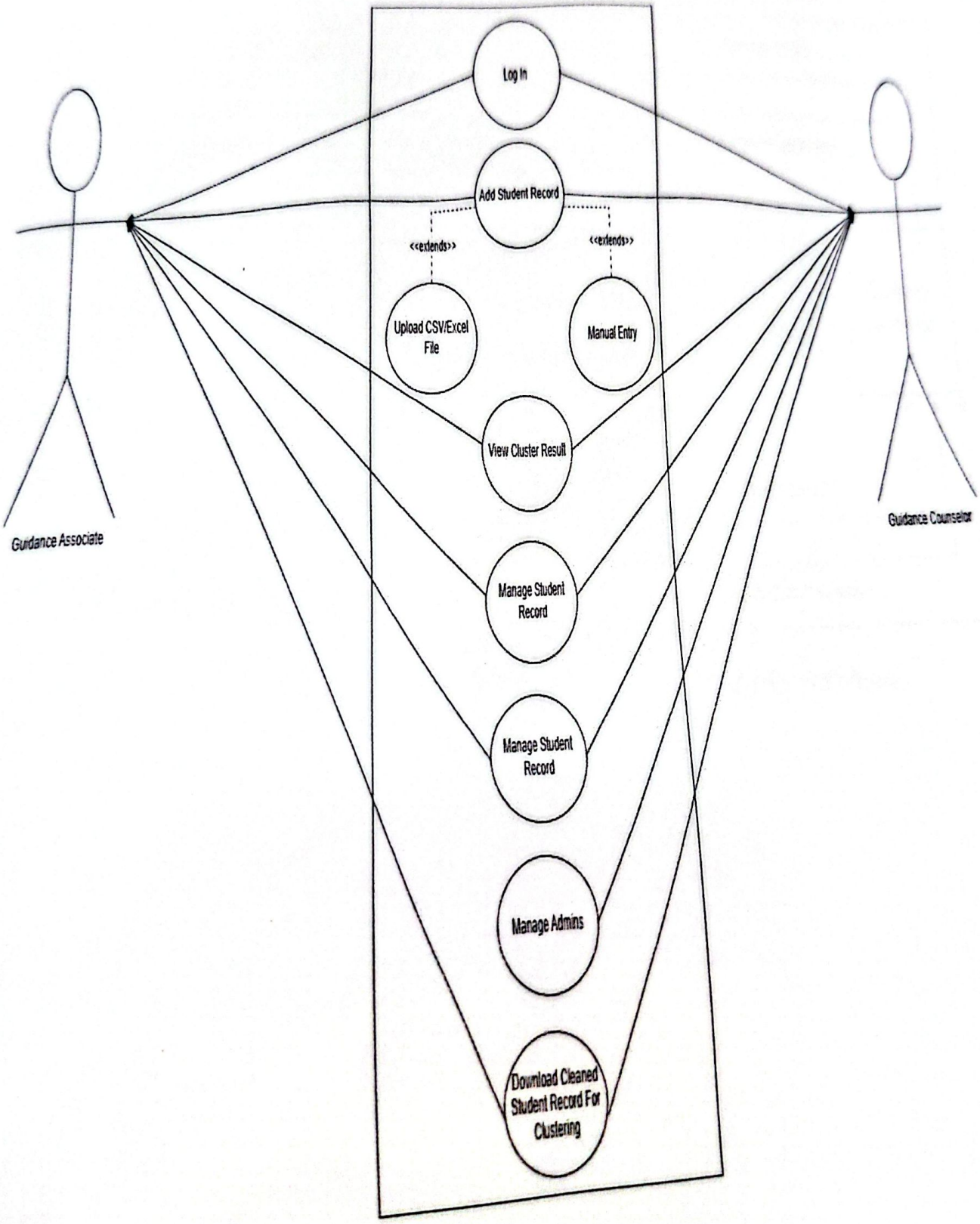
APPENDIX D

DATA FLOW DIAGRAM LEVEL 0



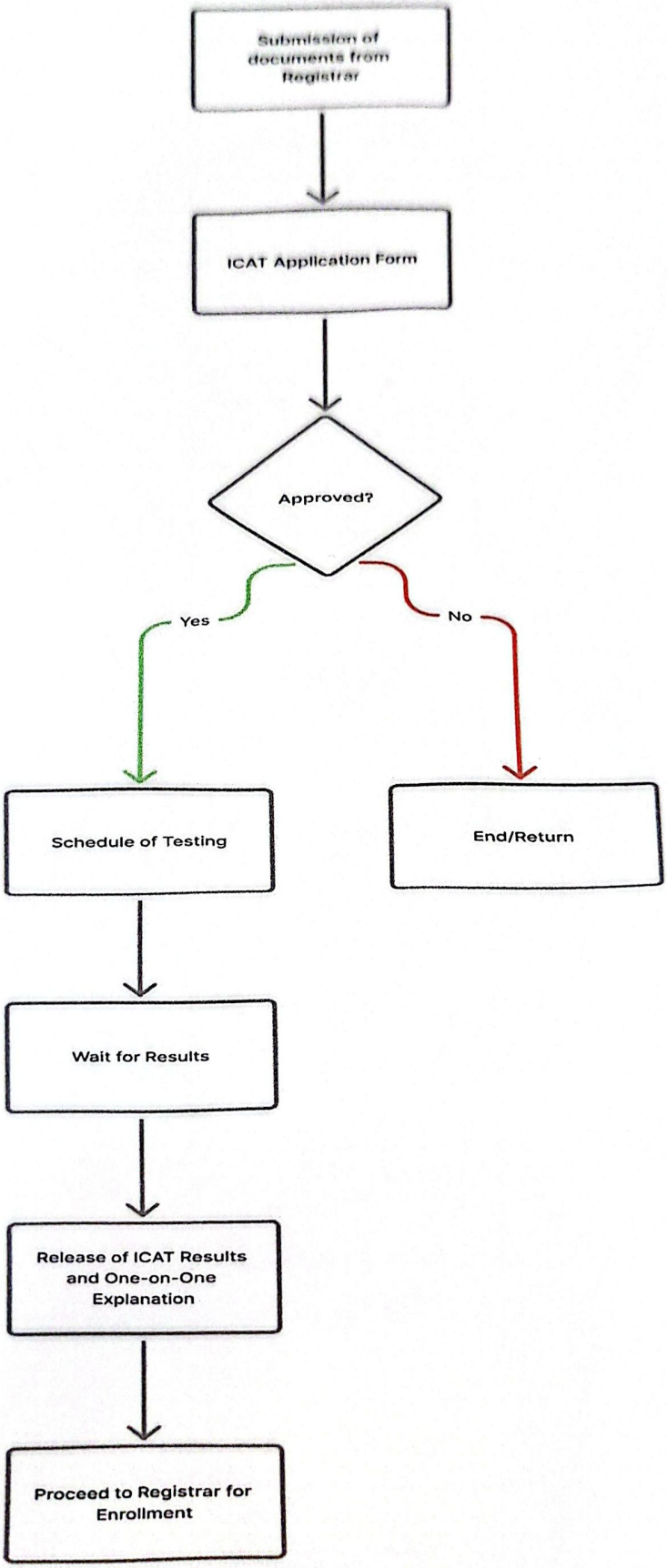
APPENDIX C

USE CASE DIAGRAM



APPENDIX B

ICAT PROCESS FLOW CHART



APPENDIX A

COMMUNICATION LETTER

Republic of the Philippines
ILOCOS SUR POLYTECHNIC STATE COLLEGE
www.ispsc.edu.ph | tagudin@ispsc.edu.ph

College of Arts and Sciences

July 2, 2025

DR. OMAR C. VALENZUELA
Guidance Counselor

Attention: **DR. OMAR C. VALENZUELA**
Guidance Counselor

Sir,

We respectfully seek your approval to conduct an initial meeting with the Guidance Counselor of our institution to discuss the current practices and challenges in handling ICAT (Ilocos Sur Polytechnic State College Admission Test) data at ISPSC Tagudin Campus. This meeting forms part of our preparatory activities for Capstone 1: Research Methods in Information Technology, under the Bachelor of Science in Information Technology program.

The insights we gather from this discussion will directly inform the development of our proposed capstone project entitled "Clustering Student ICAT Score Using Machine Learning Algorithms." The aim of this project is to develop a system that applies unsupervised machine learning techniques specifically K-Means Clustering to analyze and group students based on their ICAT scores in areas such as General Ability, Verbal Aptitude, Numerical Aptitude, Spatial Aptitude, Perceptual Aptitude, and Manual Dexterity. The results of this clustering will be used for academic profiling, course advising, and research purposes, and may also help in identifying students who may require support or who are eligible for board-related courses.

Please be assured that all information gathered will be handled with strict confidentiality and used solely for academic purposes. Any findings or outputs of this study will be shared with your office upon request.

We hope for your kind support and approval of this request.

Thank you for your time and consideration.

Sincerely yours,

AZRIEL JON M. YUKEE
Researcher

CHRISTINE LORENCE H. BONIFACIO
Researcher

ARIES F. MACABITAS
Researcher

JOHN MARK A. SALVADOR
Researcher

Noted

GEORGE VILLANUEA JR, DIT
Adviser, Capstone 1

JIM-MAR F. DELOS REYES, MIT
Program Head, BSIT

JOY G. BEA, DIT
Dean, CAS

Approved

DANIEL JUAN B. RAMIREZ, PhD
Campus Director

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Unsupervised Learning – refers to a class of machine learning procedures that discover patterns or groupings within data without predefined desired outcomes.

Web-Based System – A software application which is accessible and controllable by a user through a web browser over the internet.

Feature Selection – A process of identifying and selecting the most relevant variables to consider for model development.

ICAT – The standardized aptitude test adopted by the Ilocos Sur Polytechnic State College to gauge the cognitive and the psychomotor abilities of all incoming students.

K-Means Clustering Algorithm – An unsupervised machine learning algorithm utilized to segment data into a certain number of groups, called clusters, based on the similarity of their features.

Machine Learning – A branch of artificial intelligence that gives the ability to computers to learn from data without explicit programming and, hence, make predictions or decisions.

Modeling Phase – The CRISP-DM phase in which machine learning algorithms are utilized to analyze and interpret data.

Normalization – This is the process of scaling values in a dataset to a common range to avoid distortions caused by differences in value ranges.

Percentile Rank – The score indicating a student's relative standing among peers, expressed as a percentage.

Silhouette Analysis – A method to judge the quality of clustering based on a comparison between an object and its own cluster, versus other clusters.

SUS (System Usability Scale) – A standardized questionnaire that allows assessing system usability based on user satisfaction, efficiency, and ease of use.

GLOSSARY

Algorithm – An ordered process or set of rules for solving a particular problem or performing a computation.

K-Means Clustering – A technique of unsupervised machine learning that groups data points based on similarities for the discovery of hidden patterns or relationships within a set of data.

CRISP-DM – An acronym for Cross-Industry Standard Process for Data Mining, it is a widely used model of a systematic process in data mining and analytics. Its six stages are: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.

Data Mining – The process of analyzing large datasets to discover patterns, relationships, and useful insights.

Dataset – A collection of data that is organized and can be analyzed, possibly in tabular form with rows and columns.

Descriptive Developmental Research Design – A research design that involves description and development in mapping out the design of systems or models using observed data.

Elbow Method – It is a method adopted in the K-Means clustering to identify the number of clusters beyond which addition results in diminishing returns.

Evaluation Phase – The phase of the CRISP-DM model, during which the developed model's accuracy, validity, and usefulness are evaluated.

4. Replication of this system with larger datasets, incorporating additional factors such as academic performance, attendance, and behavioral indicators, is recommended to develop more comprehensive student profiles.
5. Wider testing across multiple ISPSC campuses is encouraged to validate system performance, usability, and effectiveness, promoting broader adoption and ensuring that the system serves as a reliable tool for academic guidance and course recommendations.

5. Usability evaluation using the System Usability Scale (SUS) indicated that the system is generally usable and user-friendly, receiving a mean score of 68.75 ("Good"). The interface is functional, and clustering results are clear, making the system an effective decision-support tool for guidance counselors.

Recommendations

Based on the findings and resulting conclusions, the following recommendations are made. These suggested acts will further enhance performance, functionality, and institutional integration of the ICAT Clustering System. The recommendations are targeted at guiding future enhancements, enabling campus-wide adoption, and supporting the system's continued effectiveness in the management and analysis of educational data.

1. ISPSC Tagudin Campus should formalize the adoption of the ICAT Clustering System in the Guidance Office to replace manual interpretation of ICAT results. This will improve accuracy, efficiency, and accessibility of student data, enabling faster report generation and better-informed academic advising.
2. The system interface and features should be continuously refined to enhance usability for both technical and non-technical users. Periodic usability testing, training sessions, and maintenance by the ISPSC IT Department are recommended to ensure data security, operational stability, and long-term sustainability.
3. Future research should explore the integration of additional machine learning algorithms, such as DBSCAN, Hierarchical Clustering, or Gaussian Mixture Models, to compare their performance with K-Means clustering.

Conclusions

This study was conducted to design, develop, and evaluate the Clustering Student ICAT Scores Using Machine Learning Algorithm system in support of ISPSC Tagudin Campus toward better management, analysis, and interpretation of the results from ICAT. Based on the research objectives, the following conclusions were drawn to end:

1. The current ICAT score interpretation process at ISPSC Tagudin Campus is still performed manually. While accurate, it is time-consuming and limits the efficient analysis of student performance trends. This highlights the need for an automated system to improve efficiency, reliability, and the usefulness of ICAT results in academic guidance.
2. The K-Means clustering algorithm was successfully implemented following the CRISP-DM framework to automatically group students based on performance across six aptitude areas. The optimal number of clusters was determined using the Elbow Method and validated through the Silhouette Score, Calinski–Harabasz Index, and Davies–Bouldin Index, confirming that the clusters were meaningful and accurately reflected student profiles.
3. The developed web-based system effectively stores ICAT scores, generates dashboards, and recommends suitable academic programs according to student profiles. This supports data-driven decision-making and enhances academic guidance processes.
4. The system has potential to assist in informed course recommendations and in assessing eligibility for board courses, providing a reliable tool for student profiling and curriculum planning.

Table 5. Level of Usability of the Clustering Student ICAT Score Using Machine Learning Algorithm

MEAN	GRADE	STANDARD DEVIATION	MIN	MAX
68.75	B	1.77	67.5	70.0

The results in Table 5 reveal that the overall mean usability score for the Clustering Student ICAT Score System was 68.75, which falls within the grade B (Good) rating according to the SUS interpretive standard. This also surpasses the benchmark average of 68, so the system adequately meets acceptable standards for usability and can be generally effective for the purpose it serves in clustering ICAT scores of students.

The calculated standard deviation of 1.77 indicates a very small dispersion among the respondents' ratings, which should mean that all users have been consistent in their perceptions of the usability of the system. The minimum score of 67.5 and the maximum score of 70.0 show that all the responses are within the "Good" range. This consistency suggests that the users-regardless of how diverse in experience and background, or inexperienced with the system-find it reliable, easy to use, and functionally coherent.

Overall, the obtained mean score implies that the Clustering Student ICAT Score System is perceived as a usable and efficient tool to carry out educational data analysis. Users could easily work their way around the system and carry out tasks related to clustering effectively. Though the results indicate a very good usability performance, a few minor enhancements in interface and user interaction would raise this rating from "Good" to "Excellent" in future versions.

system to be consistent and logically organized. Consistent color schemes, placements of buttons, and design patterns help in good user experience.

In Question 7, "I would imagine that most people would learn to use this system very quickly," both respondents agreed or strongly agreed. This result infers that the system has a low learning curve, hence users can quickly understand its operations without extended training. The said learnability is one of the most important indicators of good usability.

For question 8, "I found the system very cumbersome to use," one strongly disagreed, while another disagreed. The sum of these disagreements signifies that none of the users found the system cumbersome or tiresome to use, indicating a system design that is efficient and responsive to support smooth task execution.

For Question 9, "I felt very confident using the system," both respondents agreed in their response. In essence, this unanimous agreement will mean that users felt comfortable and self-assured while interacting with the system. Confidence usually arises when performance is sound and the interface clear and predictable.

Finally, Question 10, "I needed to learn a lot of things before I could get going with this system," both respondents disagreed. That consistent response would confirm that users found the system relatively easy to learn and did not require extensive prior knowledge. This is a reflection of strong initial usability and effective design, allowing new users to become productive quickly.

Overall, responses in Figure 6 show that users generally perceive the system as easy to use, consistent, and learnable, although some refinements in engagement and feature integration could further improve user satisfaction.

with its features. Further engagement and usefulness of the system could be improved in seeking more frequent use.

For Question 2, "I found the system unnecessarily complex," one disagreed and the other agreed. This can show that while there was much ease of navigation of the system for one user, another perceived some sense of complexity. These differing opinions may stem from varying levels of technical experience; thus, clearer instructions or simplified layouts might improve usability even for advanced users.

In Question 3, "I thought the system was easy to use," one strongly agreed, while the other agreed. Such unanimous positive feedback from the respondents indicates that the interface and functionality of the system are intuitive and user-friendly. Both users were able to complete tasks with minimal difficulty, which signifies strong usability in design.

In Question 4, "I think that I would need the support of a technical person to be able to use this system," one respondent strongly disagreed and the other disagreed. Their disagreement completely illustrates that both users felt able to operate the system themselves without any technical help. This suggests that the system is learnable and does not need specialized training.

In Question 5, "I found the various functions in this system were well integrated," one respondent agreed and the other was neutral. This shows that the components of the system are relatively cohesive, although some users still feel that there is some room for improvement in the integration of features. Continued refinement in how modules are connected could strengthen user satisfaction even more.

For Question 6, "I thought there was too much inconsistency in this system," both respondents disagreed. From this strong consensus, users found the interface of th

and standardizes the data automatically, including textual entries being transformed into numeric ones, to make the file suitable for processing by the K-Means clustering algorithm. In this way, it makes data preparation easier and ensures that the clustering outcome will remain as accurate and consistent as possible.

Usability Evaluation of the ICAT Clustering System

Figure 20 presents the distribution of user responses for each of the ten items in the System Usability Scale (SUS) questionnaire. The data reveal varying degrees of agreement across all statements, indicating differing perceptions of the system's usability.

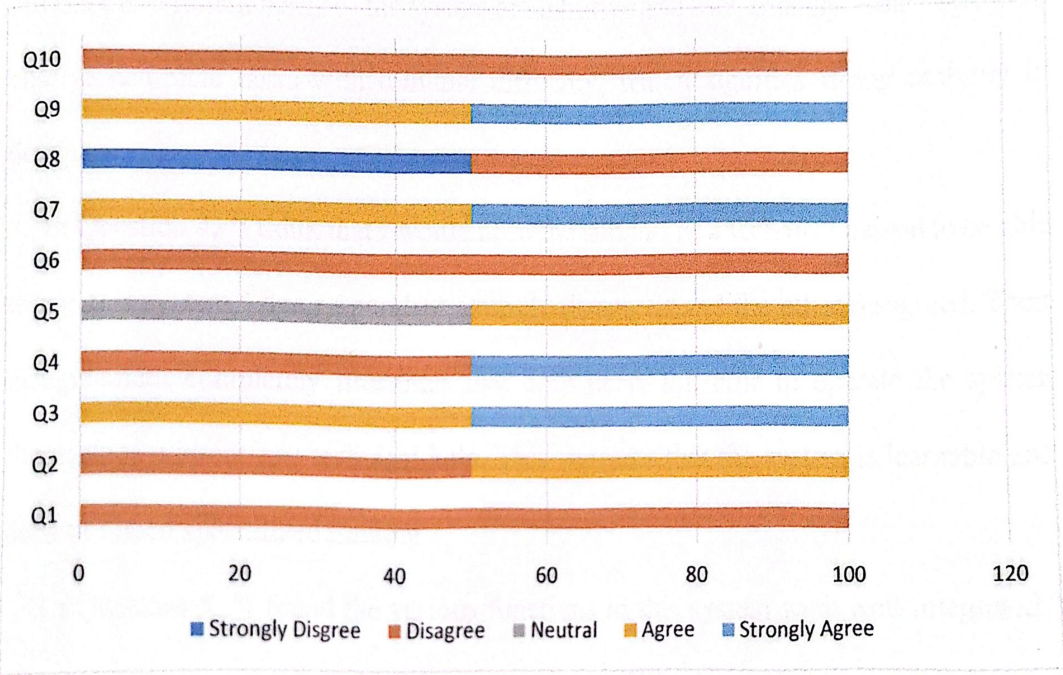


Figure 20. SUS Distribution Responses

In question 1, "I think that I would like to use this system frequently," both subjects expressed a negative view. That means users at this stage were not yet willing to use the system regularly, probably because of limited exposure or early-stage interaction

Description, and IP Address. These details help identify what happened, who did it, and where the action came from.

Each log entry indicates the date and time the action occurred precisely. The admin column identifies the user responsible-in this case, a repeating Guidance Counselor account. The Action column uses a red pill-shaped label that says "Deleted," giving a clear indication at a glance regarding the type of action that has been performed. The Target column shows what was affected; in this case, an "Admin Account." Additional information like deleted account name and email may be provided in the Description column, which provides clarity and supports proper auditing. The IP Address column to the far-right tracks where the action originated, helping find unusual or unauthorized activity.

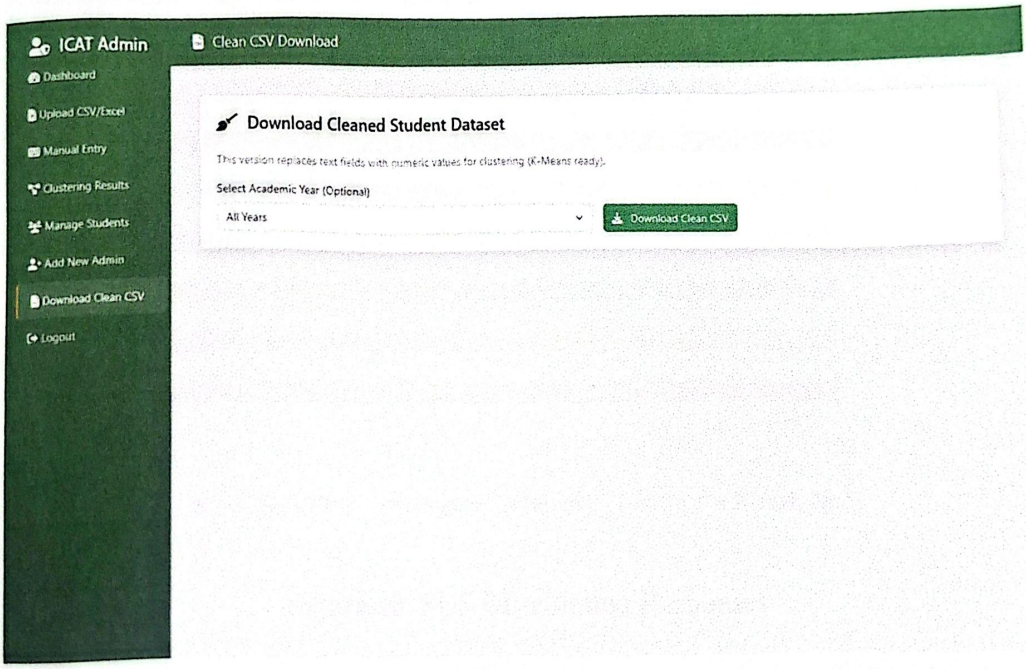


Plate 14. Download Clean CSV Page

Plate 14 presents the Download Clean CSV page, where administrators can export a preprocessed dataset that is ready for clustering. The users can select either an academic year or All Years if they want to download it all at once. The system cleans

name, role assigned, and contact email. The Position column employs colored, pill-shaped badges to visually distinguish such roles as Guidance Counselor, Guidance Associate, and Guidance Clerk. The Email column has each admin's official contact address. On the far right, the Actions column includes two strongly colored buttons for each account: a blue Edit button to update any admin information and a red Delete button to remove an account if necessary. The distinct coloration doesn't cause confusion and makes the admins' management fast and easy.

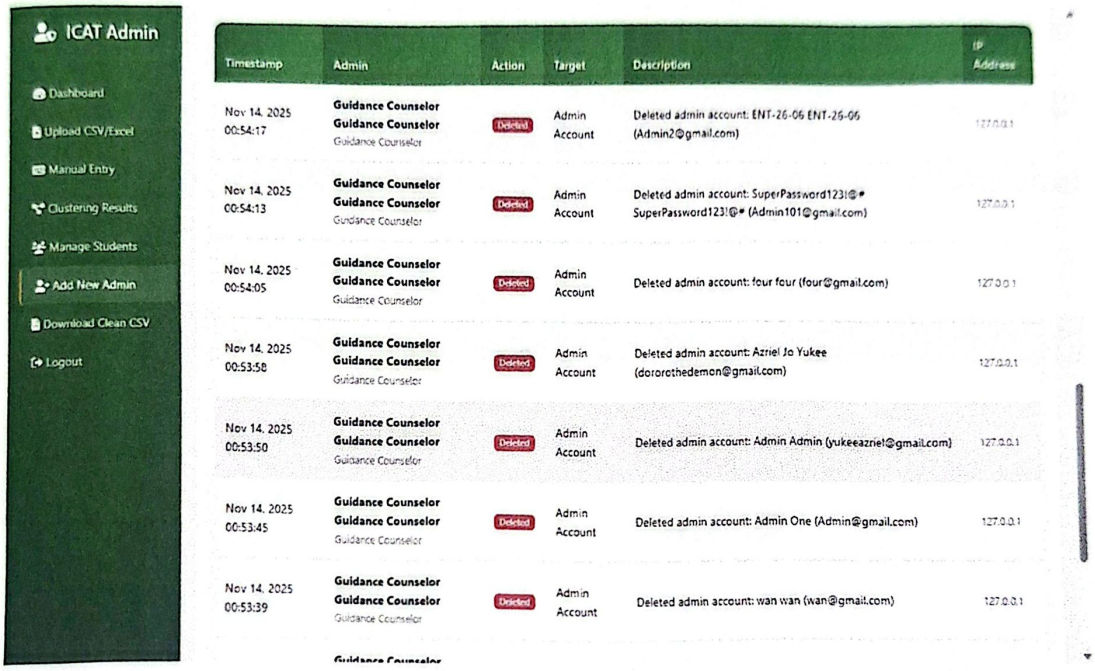


Plate 13. Audit Logs

Plate 13 shows the Audit Logs page displays all the administrative actions occurring on the ICAT system for transparency and tracking purposes. The left dark green sidebar is the same as that on every other page of the system and allows quick access to other pages such as Dashboard, Upload CSV/Excel, Manual Entry, Clustering Results, Manage Students, Add New Admin, Download Clean CSV, and Logout. Above, the main area shows a big, neatly formatted table listing every activity that has been logged. The dark green table header contains columns like Timestamp, Admin, Action, Target,

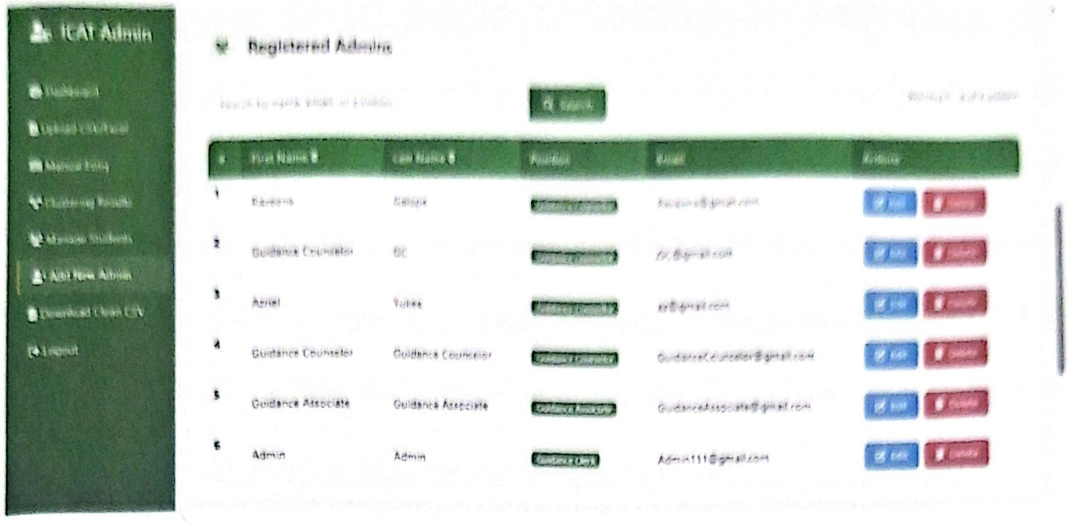


Plate 12. Registered Admins

Plate 12 shows the Registered Admins page, where all the administrative accounts in the ICAT system are displayed in a clean and organized layout. The left dark green sidebar remains visible for quick access to other modules of interest, including Dashboard, Upload CSV/Excel, Manual Entry, Clustering Results, Manage Students, Add New Admin, Download Clean CSV, and Logout.

The title Registered Admins is written at the top of the main content area with a multi-user icon next to it, showing that this page is used for managing admin accounts. Just below the header is a search bar that allows users to quickly find an admin by typing in their name, email, or even position. A Search button beside it makes filtering fast and easy. On the right side of the header, a small label indicates the current list count here displayed as "1 - 6 of 6 admins" so users know how many admin profiles are stored in the system.

On this main part of the page, a neatly designed table with a green header and columns for Number, First Name, Last Name, Position, Email, and Actions can be found. Each row represents an administrator registered under the program, shown by

CSV/Excel, Manual Entry, Clustering Results, Manage Students, Add New Admin, Download Clean CSV, and Logout. The Add New Admin option is highlighted, meaning the user is currently on the admin creation page.

The header Create New Admin at the top of the main panel, with its corresponding icon for creating a user, very clearly states the purpose of the page. This interface is divided into two sections: Personal Information and Account Security, both placed into soft, white panels that keep a clean, professional look and feel.

In the Personal Information section, the form asks for basic personal information to create a new admin account. First Name, Last Name, and Middle Name are all provided with placeholder texts that guide a person's input. A dropdown menu labeled Position allows the selection of the admin's role, such as 'Guidance Counselor.' An Email Address field, highlighted with a distinct background color, ensures that important contact information is clearly visible. All fields are well-spaced to make the form easy to follow.

Below the Account Security section is where the user provides the password for the new admin. The Password field contains a visibility toggle icon. A Confirm Password field is available to ensure the password is correct. A small note below reminds users of the password criteria-a minimum of 8 characters that should include uppercase, lowercase, a number, and a special character.

A create account button in green at the bottom submits the form to complete the registration, while the yellow Reset Form clears all the inputs if the user wants to start over. Overall, the design is intuitive and user-friendly, clearly structured to guide the administrator through adding a new account without confusion.

Number, Admin who performed the update, Field Changed, along with Old Value, New Value, and Change Type. These columns make sure that full transparency and accountability in all updates made in the system are realized.

At the top of the modal, the toggle indicator shows all system modification logs are being displayed on the page. Since no updates have yet been recorded, a friendly message in the center says, "No modification logs found", with an icon for elucidation to help administrators immediately understand that no changes exist so far.

A basic Close button positioned at the bottom right allows the user to close the modal and return to the Student Records table. The overall design is minimal, tidy, focused on the core purpose of reliably tracking changes in data, and supports the security and responsibility required in managing student records effectively.

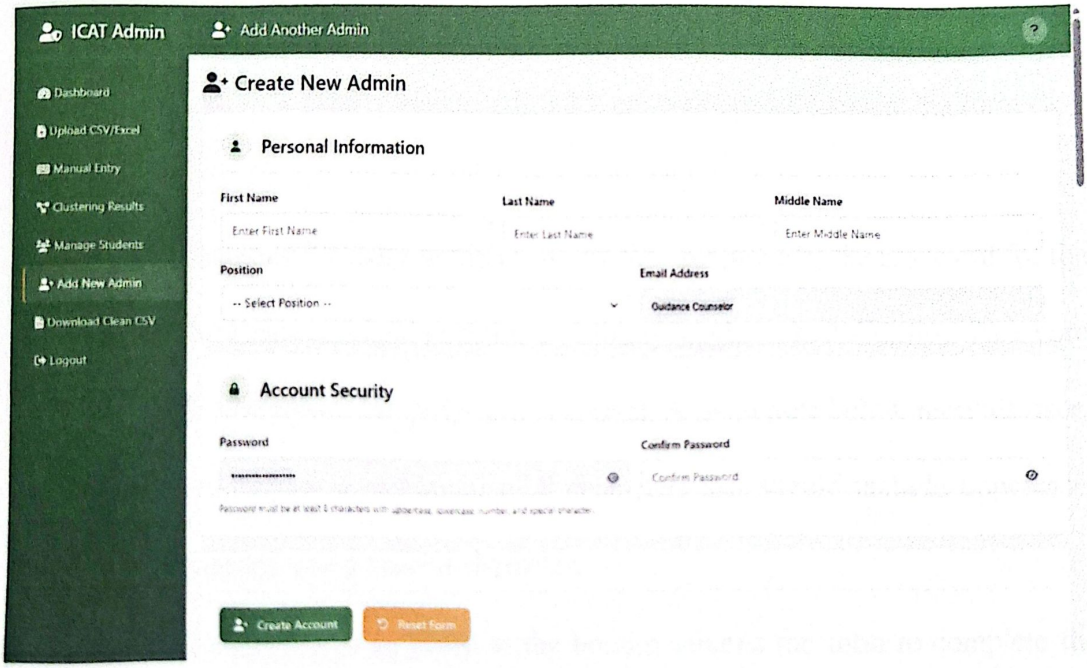


Plate 11. Add New Admin Page

Plate 11 presents the Add New Admin page, which gives a clean and clear interface for registration of the new administrative account in the ICAT system. On the left, a dark green vertical navigation panel lists options such as Dashboard, Upload

Manual Dexterity information to support the screening of academic performance. Every row also exhibits a column showing Date Taken, and two action buttons: a blue View icon for checking full student details and a yellow Edit icon for modifying records. Color-coded badges make fields such as Sex, Strand, and Course visually different, and keep the table clear and easy to navigate.

The left sidebar provides consistent navigation, allowing users to seamlessly move through modules like Dashboard, CSV Upload, Manual Entry, Clustering Results, Manage Students, Add New Admin, and Download Clean CSV. The interface in general has been designed to make large volumes of student data management easy, organized, and user-friendly.

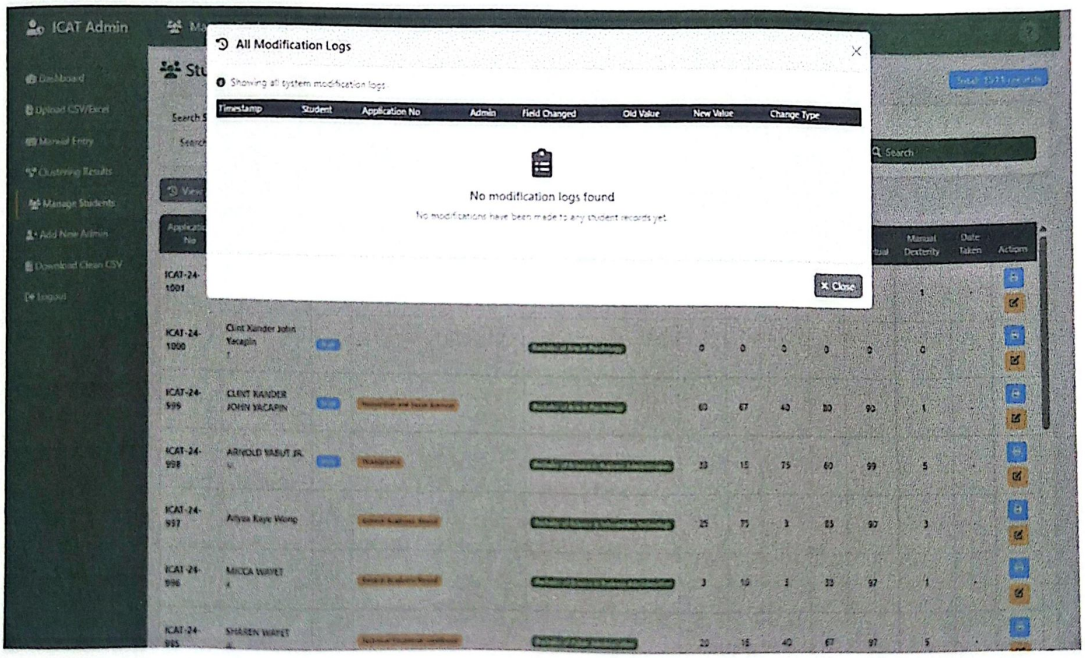


Plate 10. Modification Logs

Plate 10 presents the Modification Logs window, which is a single location for administrators in the ICAT Admin System to review all changes to student records. The modal displays a table with an easy-to-understand structure and includes columns for capturing essential audit information: Timestamp of modification, Student, Application

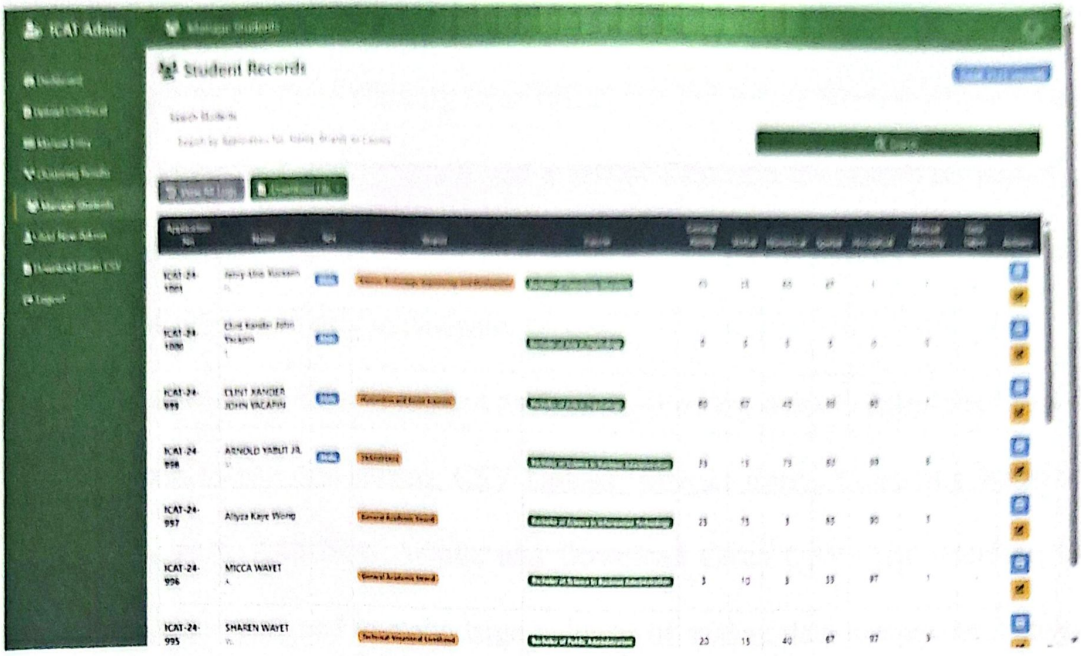


Plate 9. Manage Student Records Page

Plate 9 shows the Manage Student Records page, which serves as a focal point where all student entries can be viewed and managed in the ICAT Admin System. The top part contains a search bar where administrators will be able to perform quick searches among students by using keywords such as Application Number, Name, Strand, or Course. A green Search button is conveniently placed beside it for the execution of queries.

Below the search section, there are two action buttons: View All Logs, which allows users to access the records of the system activities, and Download File, which offers various options to export student information in different formats. On the right-hand side of the header, there is a summary badge indicating the total number of student records present in the system for the administrator's quick view.

The main section houses a well-structured table of the essential details for each student. The columns include Application Number, Name, Sex, Strand, Course, and all of the test scores on ICAT: General Ability, Verbal, Numerical, Spatial, Perceptual, and

interface displays the resulting student clusters and provides access to analytical tools such as visual analytics and pattern recognition. These tools allow administrators to see how students are grouped based on their ICAT scores and to identify noticeable patterns in academic performance. A button labeled "Open Clustering Dashboard" lets users explore the data further through interactive visualization and deeper analysis.

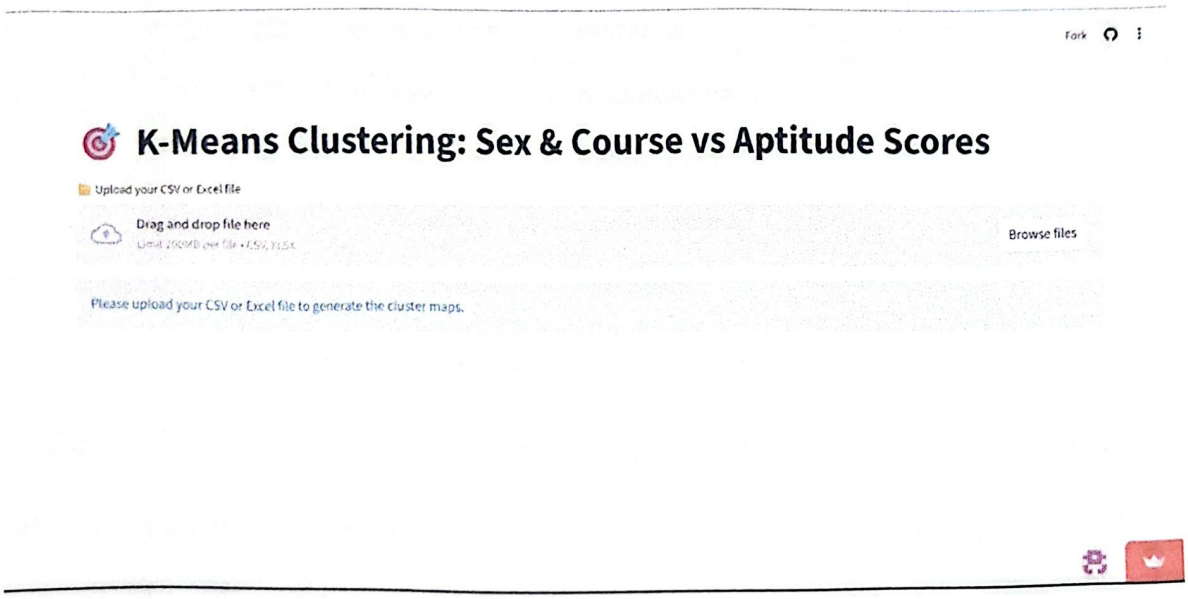


Plate 8. Clustering Results Page

Plate 8 shows the K-Means Clustering web interface, implemented using Streamlit and titled "K-Means Clustering: Sex & Course vs Aptitude Scores." This page allows users to upload CSV or Excel files containing student data in order to create cluster maps for further analysis. It includes a drag-and-drop upload area with indicated file format and size limits, as well as an option to manually browse and select a file.

Just below this section is a dropdown to select the Guidance Counselor assigned to the student. Next is the Test Scores panel, which comprises several dropdown menus where the admin will input the student's results for each aptitude category, namely General Ability, Verbal Aptitude, Numerical Aptitude, Spatial Aptitude, Perceptual Aptitude, and Manual Dexterity. The design makes use of modern card-style containers with icons for added clarity, while the form is spaced and grouped in a manner that provides smooth and intuitive data entry.

The left-side navigation sidebar enables administrators to easily switch between Dashboard, Upload CSV/Excel, Clustering Results, Manage Students, Add New Admin, Export Clean CSV, and Logout, hence providing a full-cycle administrative workflow within one consistent interface. On the whole, the page structure would facilitate accuracy, organization, and usability by staff managing student records.

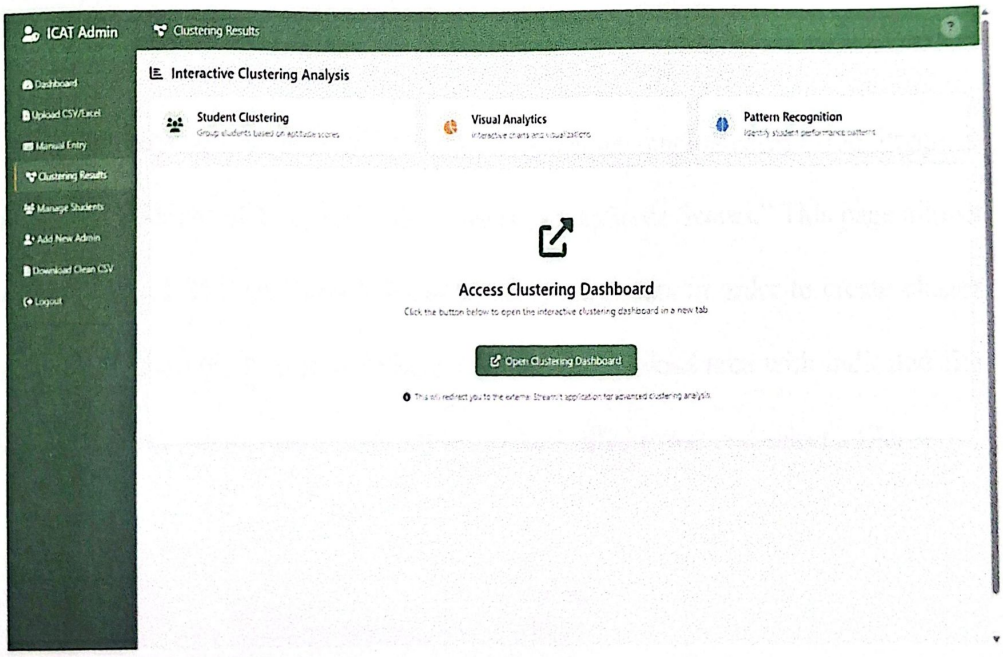


Plate 7. Access Clustering Dashboard Page

Plate 7 shows the Access Clustering Dashboard page, where the system presents the output of its machine learning process using the K-Means clustering algorithm. This

Plate 5 displays the Upload Student Data page in the ICAT Admin interface. Here, the admin can upload CSV or Excel files containing student information for clustering and analysis. The central area features a green drag-and-drop box with an upload icon, and an alternative Browse Files button. At the bottom, there is an Upload & Process button that processes the uploaded file for use within the system.

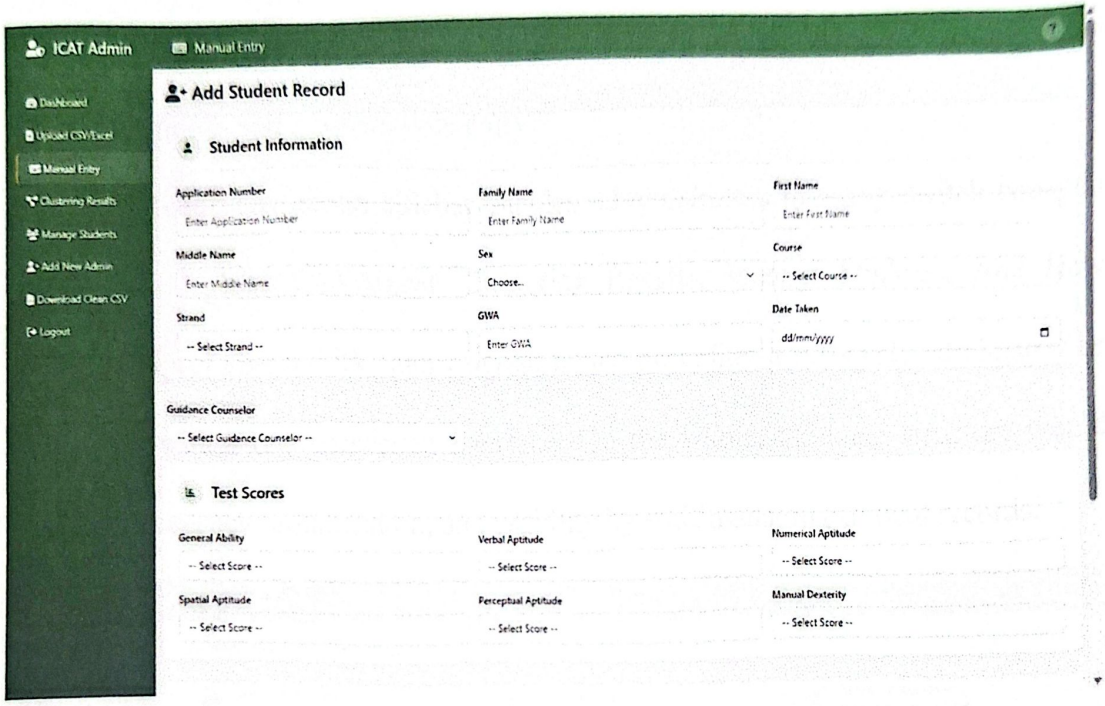


Plate 6. Manual Entry Page

Plate 6 presents a neat, organized interface for manually adding a student record to the ICAT Admin System. The complete student information appears in the form on the page. The header of the page explicitly tells the admin that they add a new student record. All the identifying personal and academic details of the student exist within the section Student Information. These are the Application Number, Family Name, First Name, and Middle Name. Also, dropdowns include Sex, Strand, Course, and a date picker for the Date Taken. An entry solely for the GWA of the student provides for recording all the information pertaining to the student.

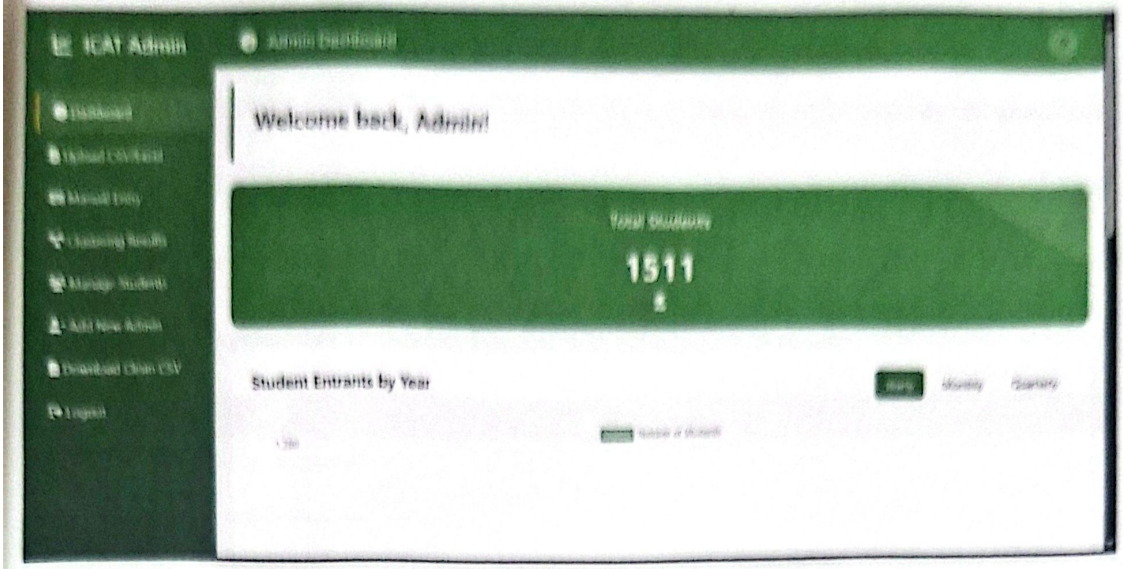


Plate 4. Admin Dashboard

Plate 4 presents the Admin Dashboard, which is the control center of the ICAT Admin system. It shows a summarized view of the key system metrics, one of them being the Total Number of Students, present at the top. There is a graphical representation called "Student Entrants by Year," where the admin can review the student trend in various time dimensions such as yearly, monthly, or quarterly. The format is clear and concise to make sure that administrators easily track key information briefly.

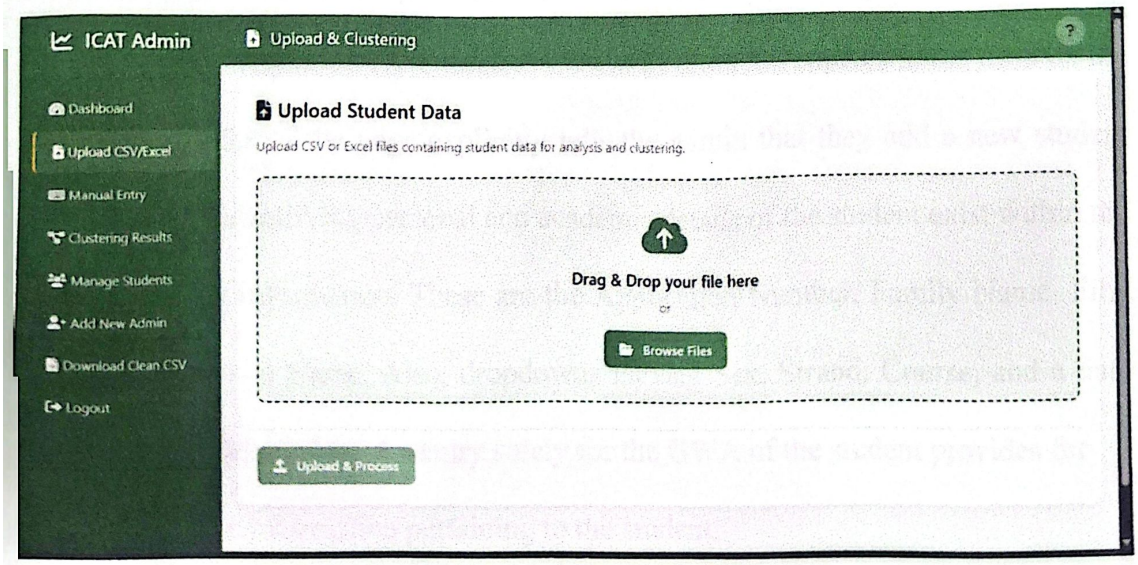


Plate 5. Upload CSV/Excel Page

A large, rounded Login button with a green-to-red gradient is positioned at the bottom of the modal for easy viewing and access by the user. On the partially visible home page behind the modal, the system title "Intelligent Student Clustering & Analytics," its introductory message, and feature cards for Bulk Data Import, Manual Entry, and K-Means Clustering orient the user within the interface.

Overall, the login modal provides a focused, accessible, and secure point of entry for system administrators while preserving the context of the main interface.

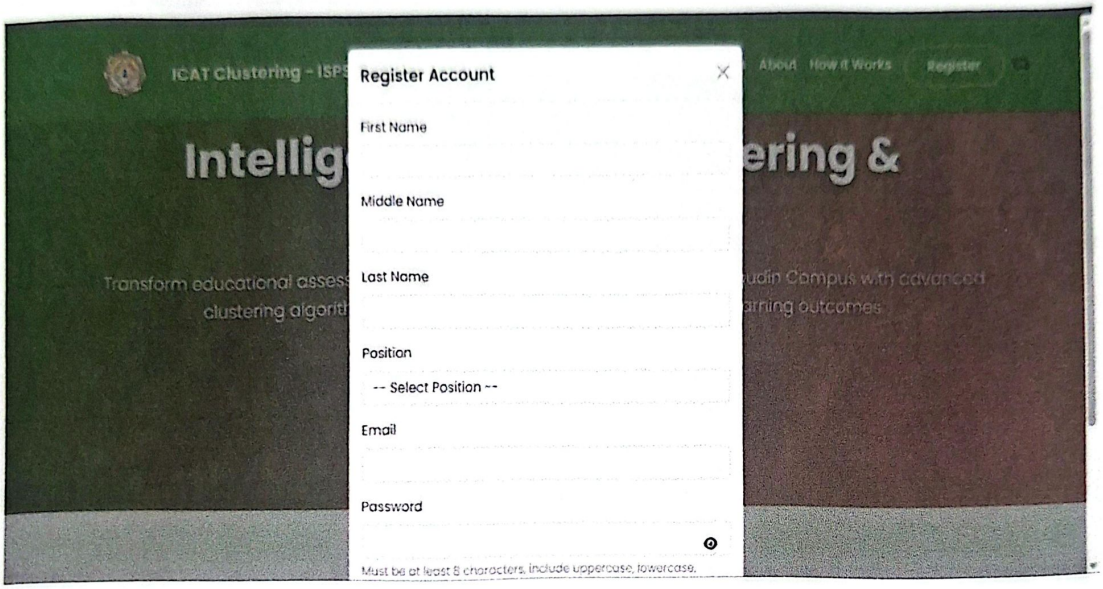


Plate 3. Registration Form

Plate 3 shows the Registration Form asks for the most basic information of a user, which includes First Name, Middle Name, Last Name, Position, and Email. It also has a Password field with details on password requirements below the password input field. At the bottom, there is a large, accent-colored Register button for users to submit their registration details.

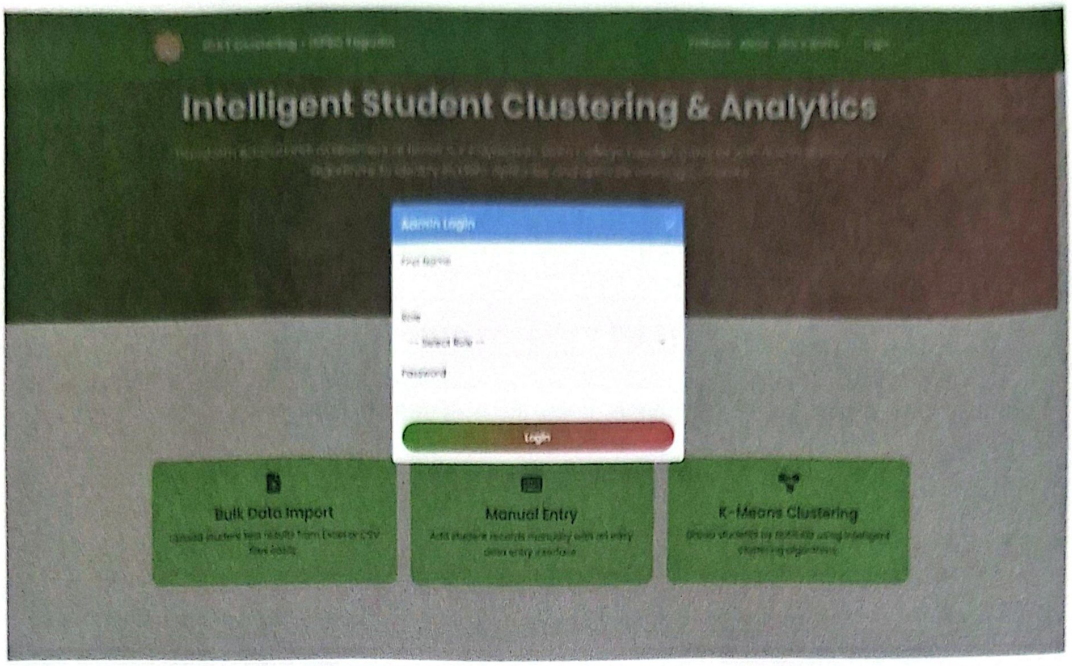


Plate 2. Login Page

Plate 2 shows the Login Interface of the ICAT Clustering System is positioned as a pop-up modal in the middle and overlaid on top of the home page to keep the user's attention on the authentication process without leaving the main screen. The underlying page is faded to subtly direct the user's attention to the form while still being aware of the general structure of the system.

At the top of the modal, there's a large blue header with "Admin Login" written on it, hinting pretty clearly that access is restricted to those with some sort of clearance. A small "X" icon positioned on the upper right provides a quick option for closing the modal and returning to the view of the homepage.

The form section includes three well-structured input elements: one field for entering the First Name of an administrator, another one for selecting their Role inside the institution from a dropdown menu, and a secure password field. Every input element is placed with proper spacing to maintain readability and lower possible encoding errors, thus making a login smooth.

Subtle background imagery, such as data charts and an academic setting, creates a modern and data-driven atmosphere without overwhelming the layout. A gold "Learn More" button is centrally placed to guide users who would want to explore the system further. As the user scrolls down, this interface transitions into the System Features section. This section briefly outlines how the platform supports efficient analysis and clustering of student aptitude data.

Three feature cards, each describing a key functionality of the system, are presented in a column below the introductory text. The “Bulk Data Import” feature allows users to upload multiple student records simultaneously using Excel or CSV files. The “Manual Entry” feature allows for the encoding of individual student information through a user-friendly input form. Meanwhile, the “K-Means Clustering” feature mentions the system’s ability to automatically segment students, based on aptitude scores, through intelligent clustering. All the feature cards are styled with clean green designs and straightforward icons to support an organized, easy-to-understand visual structure. In all, the Landing Page offers an easy, smooth introduction to the ICAT Clustering System, accessible yet professional and institutional.

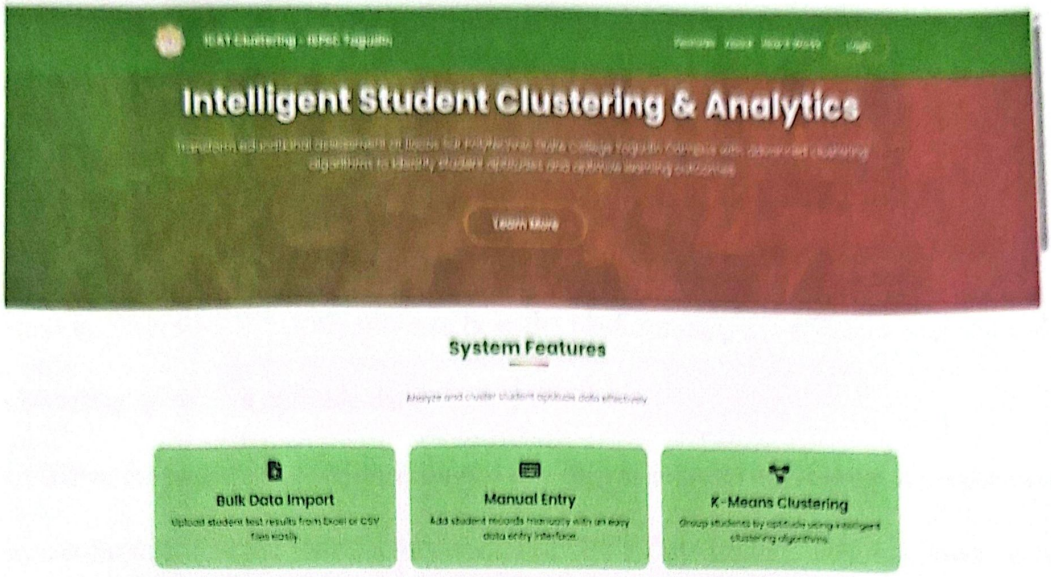


Plate 1. Landing Page

Plate 1 shows the Landing Page of the ICAT Clustering System, the main entry interface to the system-is structured in such a way that it provides clear and organized information for the users. Serving as a point of first contact, the page is designed to be simple and inviting, and highly navigable, to allow users to quickly get acquainted with what the system is about and what can be done with it.

A green header at the top of the interface displays the ISPSC logo, along with a title saying "ICAT Clustering – ISPSC Tagudin," establishing the official association of the system with the institution. On the right side of this header are the main navigation options: Features, About, How It Works, and a clearly visible Login button that allows users to quickly find major sections of the application or go directly to account access.

Below the header is a wide banner section that introduces the core identity of the system. The title "Intelligent Student Clustering & Analytics" is in bold and striking format, while a short description is attached, highlighting the system's role in improving educational assessment at ISPSC Tagudin through modern clustering algorithms.

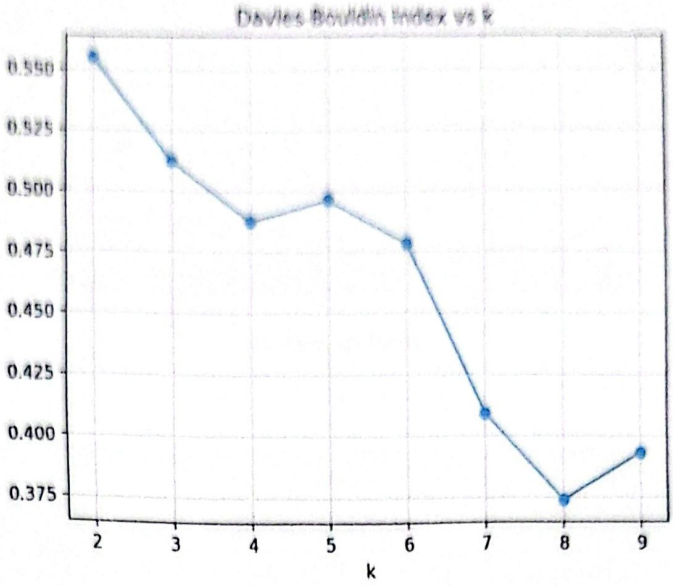


Figure 19. Davies-Bouldin Index vs k

Development of the Web-Based ICAT Clustering System

The system was developed to automate the management and clustering of ICAT scores. It integrates a machine learning algorithm that classifies students into clusters based on performance in six aptitude areas: General Ability, Verbal Aptitude, Numerical Aptitude, Spatial Aptitude, Perceptual Aptitude, and Manual Dexterity. The system consists of several major components that function together to streamline uploading, managing, analyzing, and visualizing student data.

The following plates illustrate the main interfaces and functionalities of the system:

clustering structure, reinforcing the appropriateness of the selected number of clusters.

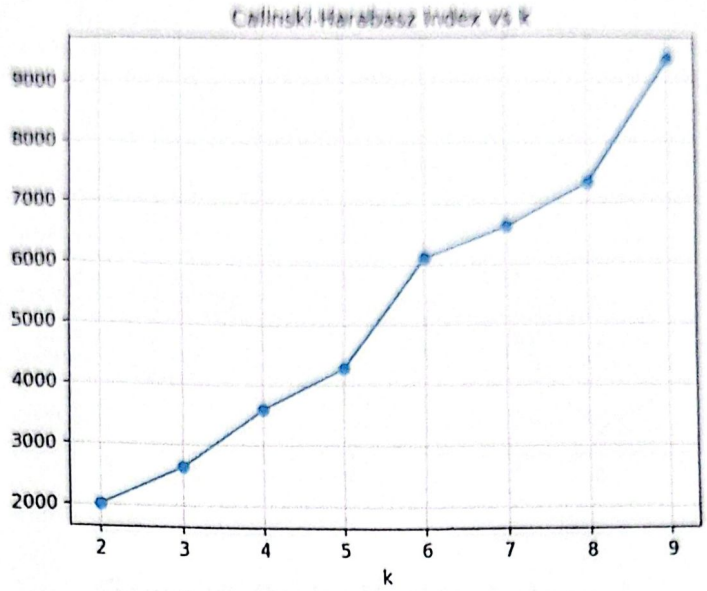


Figure 18. Calinski-Harabasz Index vz k

The Davies–Bouldin Index was also computed to measure the average similarity between each cluster and its most similar neighboring cluster. As displayed in the Davies–Bouldin Index graph in Figure X below, the model yielded a low value of 0.488, signifying low inter-cluster similarity. Since lower Davies–Bouldin values correspond to better clustering performance, this result confirms that the identified clusters were distinct and well isolated.

Performance Evaluation of the K-Means Clustering Algorithm

To assess the quality of the clustering results across different values of k, three commonly used internal cluster validation metrics were examined. The Silhouette Coefficient was used to evaluate how well each student record fit within its assigned cluster compared to other clusters. As shown in the Silhouette Score graph presented in Figure 17 below, the model achieved a Silhouette value of 0.648 at k = 4, indicating high intra-cluster similarity and clear separation between clusters. This result suggests that the formed clusters were well defined with minimal overlap among student groupings.

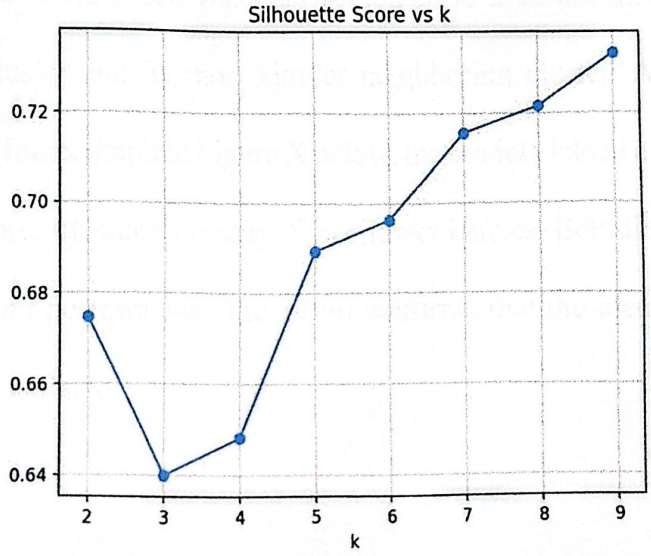


Figure 17. Silhouette Score vs k

The effectiveness of the clustering configuration was further supported by the Calinski–Harabasz Index, which measures the ratio of between-cluster variance to within-cluster variance. As illustrated in the Calinski–Harabasz Index graph shown in Figure 18 below, the clustering model produced a high value of 3,596.34 at k = 4, reflecting compact and well-separated clusters. Higher index values indicate improved

```
        # ---- Helper Function for Clustering and Plotting ----
        def cluster_and_plot(x_col, y_col, title, y_label):  12 usages
            # ==========================
            # 🔍 AUTO-DETECT BEST NUMBER OF K (Elbow Method)
            # ==========================
            inertias = []
            K_range = range(2, 10)

            for k in K_range:
                kmeans_test = KMeans(n_clusters=k, random_state=42, n_init=10)
                kmeans_test.fit(df[[y_col, x_col]])
                inertias.append(kmeans_test.inertia_)

            # ---- Determine the "Elbow" point automatically ----
            knee = KneeLocator(K_range, inertias, curve="convex", direction="decreasing")
            best_k = knee.knee if knee.knee else 3  # Default to 3 if no clear elbow
```

Figure 15. K-Means clustering with automatic k selection and evaluation

Finally, the clustering function was executed across multiple ICAT aptitude scores using sex and course as grouping variables. This enabled comprehensive analysis of student performance patterns based on demographic and academic characteristics. The execution of clustering scenarios is presented in Figure 16.

```
        # ==========================
        # 📊 SEX VS APTITUDE PLOTS
        # ==========================
        st.header("📌 SEX vs Aptitude Scores")
        cluster_and_plot( x_col: "general_ability",  y_col: "sex",  title: "K-Means Cluster Map (Sex vs General Ability)",  y_label: "Sex (1=Female, 2=Male)")
        cluster_and_plot( x_col: "verbal_aptitude",  y_col: "sex",  title: "K-Means Cluster Map (Sex vs Verbal Aptitude)",  y_label: "Sex (1=Female, 2=Male)")
        cluster_and_plot( x_col: "numerical_aptitude",  y_col: "sex",  title: "K-Means Cluster Map (Sex vs Numerical Aptitude)",  y_label: "Sex (1=Female, 2=Male)")
        cluster_and_plot( x_col: "spatial_aptitude",  y_col: "sex",  title: "K-Means Cluster Map (Sex vs Spatial Aptitude)",  y_label: "Sex (1=Female, 2=Male)")
        cluster_and_plot( x_col: "perceptual_aptitude",  y_col: "sex",  title: "K-Means Cluster Map (Sex vs Perceptual Aptitude)",  y_label: "Sex (1=Female, 2=Male)")
        cluster_and_plot( x_col: "manual_dexterity",  y_col: "sex",  title: "K-Means Cluster Map (Sex vs Manual Dexterity)",  y_label: "Sex (1=Female, 2=Male)")

        # ==========================
        # 📚 COURSE VS APTITUDE PLOTS
        # ==========================
        st.header("📌 COURSE vs Aptitude Scores")
        cluster_and_plot( x_col: "general_ability",  y_col: "course",  title: "K-Means Cluster Map (Course vs General Ability)",  y_label: "Course Code")
        cluster_and_plot( x_col: "verbal_aptitude",  y_col: "course",  title: "K-Means Cluster Map (Course vs Verbal Aptitude)",  y_label: "Course Code")
        cluster_and_plot( x_col: "numerical_aptitude",  y_col: "course",  title: "K-Means Cluster Map (Course vs Numerical Aptitude)",  y_label: "Course Code")
        cluster_and_plot( x_col: "spatial_aptitude",  y_col: "course",  title: "K-Means Cluster Map (Course vs Spatial Aptitude)",  y_label: "Course Code")
        cluster_and_plot( x_col: "perceptual_aptitude",  y_col: "course",  title: "K-Means Cluster Map (Course vs Perceptual Aptitude)",  y_label: "Course Code")
        cluster_and_plot( x_col: "manual_dexterity",  y_col: "course",  title: "K-Means Cluster Map (Course vs Manual Dexterity)",  y_label: "Course Code")

    else:
        st.info("Please upload your CSV or Excel file to generate the cluster maps.")
```

Figure 16. Execution of clustering across sex and course variables

An automated insight generation function was developed to translate numerical cluster results into meaningful textual interpretations. This function evaluated demographic dominance and compared cluster mean aptitude scores with the overall dataset mean to generate interpretable insights for each cluster. The implementation of the insight generator is shown in Figure 14.

```
# ... Insight Generator ...
def generate_insight(x_col, y_col, avg_x, avg_y, overall_mean_x):  1 usage
    if y_col == "sex":
        y_label = "Female" if avg_y < 1.5 else "Male"
        demographic_insight = f"This cluster is composed mostly of **{y_label}** students."
    elif y_col == "course":
        closest_code = min(course_mapping.keys(), key=lambda c: abs(c - avg_y))
        course_name = course_mapping[closest_code]
        demographic_insight = f"This cluster is composed mostly of students enrolled in **{course_name}**."
    else:
        demographic_insight = f"The average value for {y_col} is {avg_y:.2f}."

    if avg_x > overall_mean_x:
        ability_insight = f"The average **{x_col.replace('_',' ')}** score (**{avg_x:.2f}**) is **higher** than the overall mean."
    else:
        ability_insight = f"The average **{x_col.replace('_',' ')}** score (**{avg_x:.2f}**) is **lower** than the overall mean."

    overall = f"""
    - 📌 **Demographics:** {demographic_insight}
    - 📊 **Performance Trend:** {ability_insight}
    - 🧠 **Interpretation:** This suggests that **{y_label if y_col=='sex' else course_name if y_col=='course' else y_col}** students
      tend to have { 'stronger' if avg_x > overall_mean_x else 'weaker' } performance in
      **{x_col.replace('_',' ')}** aptitude compared to other groups.
    """
    return overall
```

Figure 14. Automated insight generation function

The clustering process was encapsulated in a helper function that performed K-Means clustering, automatically determined the optimal number of clusters using the elbow method, evaluated clustering quality using standard metrics, and generated visual outputs. This function formed the core analytical component of the system. The clustering and evaluation implementation is shown in Figure 15.

```
# ---- Required columns ----
required_cols = [
    "sex",
    "general_ability",
    "verbal_aptitude",
    "numerical_aptitude",
    "spatial_aptitude",
    "perceptual_aptitude",
    "manual_dexterity",
    "course"
]
missing = [c for c in required_cols if c not in df.columns]
if missing:
    st.error(f"❌ Missing columns: {', '.join(missing)}")
    st.write("Your file contains:", list(df.columns))
    st.stop()
```

Figure 11. Required column validation

Records with missing values in critical fields were removed to improve the reliability of clustering outcomes. Additionally, the course attribute was converted into an integer data type to ensure compatibility with numerical clustering operations. The handling of missing values and data type conversion is shown in Figure 12.

```
# ---- Drop rows with missing values ----
df = df.dropna(subset=required_cols)

# ---- Convert Course to int ----
df["course"] = df["course"].astype(int)
```

Figure 12. Removal of missing values and course type conversion

To provide users with a clear reference for interpreting clustering results, the course code mapping was displayed in a tabular format within the system interface. This table presentation is illustrated in Figure 13.

```
# ---- Show Course Mapping Table ----
st.subheader("📘 Course Code Mapping")
st.table(pd.DataFrame(list(course_mapping.items()), columns=["Code", "Course"]))
```

Figure 13. Display of course mapping table

in Figure 9.

```
if uploaded_file is not None:
    # ---- Read file ----
    try:
        if uploaded_file.name.endswith('.csv'):
            df = pd.read_csv(uploaded_file)
        else:
            df = pd.read_excel(uploaded_file, engine='openpyxl')
    except Exception as e:
        st.error(f"❌ Error reading file: {e}")
        st.stop()
```

Figure 9. Reading uploaded dataset with error handling

Once the dataset was successfully loaded, preprocessing was performed to standardize column names. This included trimming extra spaces, converting text to lowercase, and replacing spaces with underscores. These steps were necessary to maintain consistency and prevent errors during variable referencing in the clustering process. The column standardization procedure is presented in Figure 10.

```
# ---- Clean column names ----
df.columns = df.columns.str.strip().str.replace(" ", "_").str.lower()
```

Figure 10. Column name standardization process

The system validated the presence of all required attributes necessary for clustering, including demographic variables and ICAT aptitude scores. If any required column was missing, the system displayed an error message and stopped further execution to ensure data integrity and analytical accuracy. This validation process is shown in Figure 11.

into their corresponding academic program names. This mapping was essential for improving the interpretability of clustering results, particularly during automated insight generation and demographic analysis based on academic programs. The course mapping definition is shown in Figure 7.

```
# ---- Fixed Course Mapping ----
course_mapping = {
    1: 'Bachelor of Secondary Education',
    2: 'Bachelor of Elementary Education',
    3: 'Bachelor of Physical Education',
    4: 'Bachelor of Science in Business Administration',
    5: 'Bachelor of Science in Mathematics',
    6: 'Bachelor of Arts in English Language',
    7: 'Bachelor of Arts in Psychology',
    8: 'Bachelor of Arts in Social Science',
    9: 'Bachelor of Science in Entrepreneurship',
    10: 'Bachelor of Science in Information Technology',
    11: 'Bachelor of Public Administration'
}
```

Figure 7. Course code to academic program mapping

The system allowed users to upload datasets through a file uploader that supports both CSV and Excel file formats. This mechanism ensured flexibility in data input and compatibility with commonly used data storage formats. The file upload implementation is illustrated in Figure 8.

```
# ---- File Upload ----
uploaded_file = st.file_uploader("📂 Upload your CSV or Excel file", type=["csv", "xlsx"])
```

Figure 8. Dataset file uploader implementation

After a dataset was uploaded, the system automatically identified the file format and oaded the data using Pandas. A try–except structure was implemented to handle ossible file-reading errors, ensuring that the system terminated safely when invalid or orrupted files were provided. This data loading and error-handling process is shown

dataset upload and result presentation. Pandas was employed for data loading and preprocessing, while Matplotlib was used to generate graphical visualizations such as elbow plots and scatter plots. The K-Means clustering algorithm and cluster evaluation metrics were implemented using the Scikit-learn library. Additionally, the KneeLocator function from the kneed library was applied to automatically identify the optimal number of clusters using the elbow method. The implementation of these libraries is shown in Figure 5.

final_def.py

```
import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, calinski_harabasz_score, davies_bouldin_score
from kneed import KneeLocator
```

Figure 5. Importation of required Python libraries

To ensure a user-friendly and visually appropriate interface for data analysis, the system page was configured using Streamlit's page setup function. A wide layout was selected to accommodate multiple visualizations, and a descriptive title was displayed to clearly indicate the purpose of the system. This configuration allowed users to easily interpret clustering outputs and evaluation metrics. The page configuration and title display are presented in Figure 6.

```
# ---- Streamlit Page Setup ----
st.set_page_config(page_title="K-Means Clustering", layout="wide")
st.title("Clustering Student ICAT Score Using Machine Learning Algorithm")
```

Figure 6. Streamlit page configuration and system title

A fixed course code mapping was defined to convert numerical course identifiers

Development of the K-Means Clustering Model for ICAT Results

In this machine learning analysis, the ICAT dataset was first collected from the Guidance Office to ensure the use of official and validated student assessment records. Prior to model development, the dataset underwent data cleaning and preprocessing to address missing values, remove inconsistencies, and prepare the data for clustering analysis. After the ICAT dataset was prepared, it was processed using the K-Means clustering algorithm. To determine the most appropriate number of clusters, the Elbow Method was applied by evaluating the within-cluster sum of squares across varying cluster counts. This procedure enabled the identification of an optimal balance between model simplicity and clustering performance. As illustrated in Figure 4, the results indicated that four (4) clusters provided the most suitable structure for grouping students based on similarities in their ICAT aptitude profiles.

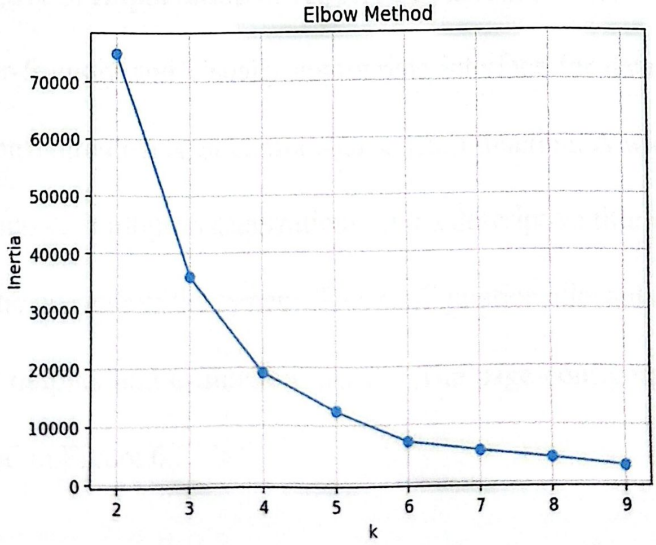


Figure 4. Elbow Method

The development of the ICAT clustering system required the integration of several ython libraries to support data processing, machine learning, visualization, and web- ased interaction. Streamlit was utilized to create an interactive web interface for

Once the examination was completed, students waited for the release of their ICAT results. These results were not immediately used for enrollment; instead, they served as the basis for further evaluation and guidance. The Guidance Office handled all test-related interpretations to ensure accuracy, confidentiality, and proper use of the assessment outcomes.

Using the official ICAT conversion table, the Guidance Office converted students' raw scores from the six aptitude areas General Ability, Verbal Aptitude, Numerical Aptitude, Spatial Aptitude, Perceptual Aptitude, and Manual Dexterity into standardized stanine scores and percentile ranks. These percentiles indicate a student's relative standing among test-takers and became the primary basis for determining academic suitability and course recommendations. Due to institutional policy and test security guidelines, the specific computation and conversion procedures were not disclosed publicly and were handled exclusively by authorized guidance personnel.

After the conversion and interpretation of scores, students underwent a one-on-one counseling session with a guidance counselor. During this session, the counselor explained the meaning of the student's percentile results, discussed strengths and areas for improvement, and recommended appropriate academic programs based on the student's aptitude profile. Only after this individualized counseling process did students proceed to the Registrar's Office for enrollment, ensuring that course selection was guided by both standardized assessment results and professional judgment.

Chapter 3

RESULTS AND DISCUSSION

This chapter presents the findings of the study and provides a comprehensive discussion of the research methods employed in developing the system entitled "Clustering Student ICAT Scores Using Machine Learning Algorithm." The discussion examines the existing process of handling ICAT data at ISPSC Tagudin Campus and presents the key features of the developed system. It further discusses the implementation of the K-Means clustering algorithm using machine learning techniques, the validation of clustering accuracy through established evaluation metrics, and the results of the system's usability assessment. Plates of the system are included to illustrate its major functionalities and user interface design. Finally, this chapter concludes with insights drawn from the findings and provides recommendations for future system enhancements.

Findings

The following are the findings related to the research questions that guided the study:

Analysis of the Existing ICAT Score Interpretation Method

At ISPSC Tagudin Campus, the interpretation of ICAT (Institutional College Admission Test) results followed a structured and systematic process prior to the development of the automated system. The procedure began with the submission of required documents from the Registrar's Office to the Guidance Office. Students then accomplished the ICAT application form, which was reviewed and approved by guidance personnel. After approval, students were scheduled for the ICAT examination and took the test on the assigned date.

functional yet requires further refinement to optimize usability. Meanwhile, scores from 38–51 are rated D (Poor), and those below 38 receive an F (Awful) grade, both indicating that substantial improvements are needed to ensure satisfactory user experience.

By integrating both quantitative measures from the SUS results and qualitative insights from user feedback, the study provided a comprehensive evaluation of the clustering system's usability and effectiveness. This dual approach ensured that the system's overall performance was assessed not only through statistical data but also through the actual experiences and perceptions of its users, resulting in a well-balanced and data-driven assessment of system quality.

Finally, to evaluate the usability and effectiveness of the developed system, quantitative data from close-ended System Usability Scale (SUS) items and post-use survey questionnaires were analyzed using descriptive statistics, including mean, frequency, and percentage. In addition, informal qualitative feedback gathered during the system presentation to the Guidance Associate and the Guidance Counselor was reviewed to supplement the quantitative results, providing additional insights into system clarity, usability, and practical relevance.

Table 4. Adjective Rating on the Level of Usability of the ICAT Clustering System

SUS Score	Grade	Adjective Rating
>80.3	A	Excellent
–80.3	B	Good
51–68	C	OK
38–51	D	Poor
<38	F	Awful

This table presents the usability grading scale based on the System Usability Scale SUS) scores. Each score range corresponds to a specific letter grade and descriptive ıting. A SUS score of 80.3 and above receives a grade of A (Excellent), indicating that ıe system is highly user-friendly and performs exceptionally well in meeting user ;pectations. Scores between 68 and 80.3 are graded as B (Good), signifying that the ·stem is generally efficient but may still benefit from minor enhancements. Scores ithin the 51–68 range correspond to a C (OK) rating, suggesting that the system is

Data Analysis

The data gathered in this study were analyzed in accordance with the specific objectives of the research. To examine the current process of interpreting and utilizing ICAT scores at ISPSC Tagudin Campus, qualitative data were obtained through document review and structured discussions with the Guidance Associate and the Guidance Counselor. These data were analyzed to identify existing practices, procedures, and limitations in the utilization of ICAT results, establishing a baseline for system development.

To analyze the ICAT scores of students, the K-Means clustering algorithm was applied to the six aptitude components measured by the ICAT. Prior to clustering, the data was prepared and normalized to ensure consistency across variables. The algorithm grouped students based on similarities in their aptitude profiles, enabling the identification of performance-based categories that reflect dominant cognitive and practical abilities.

To evaluate the accuracy and performance of the K-Means clustering model, internal cluster validation metrics were utilized, namely the Silhouette Score, Calinski–Harabasz Index, and Davies–Bouldin Index. These metrics were analyzed to assess cluster cohesion, separation, and overlap, determining the reliability and validity of the generated clusters.

To analyze the functionality of the developed web-based system, system-generated data such as stored ICAT records, cluster assignments, academic program recommendations, and dashboard visualizations were examined. The analysis focused on verifying the correctness, consistency, and accuracy of system outputs in supporting student profiling and academic guidance.

evaluators. They were selected as respondents because they are the intended and actual users of the system, responsible for interpreting ICAT results and utilizing the generated insights for academic guidance and decision-making. Limiting the evaluation to these roles ensured that the usability assessment reflected practical, real-world usage and provided valid feedback on the system's effectiveness, efficiency, and suitability for its intended purpose.

Research Instruments

To evaluate the usability and effectiveness of the clustering system, this study used the System Usability Scale (SUS) as its primary research instrument. The SUS is a standardized and widely accepted tool for measuring system usability, chosen because it provides a quick and reliable assessment of user-friendliness, efficiency, and overall user experience important factors for evaluating an educational decision-support system such as the ICAT Clustering System. The SUS was administered to the Guidance Counselor and the Guidance Clerk, who are the system's only intended end users. This is methodologically acceptable, even as the number of respondents is small, because SUS measures usability from actual users rather than a broad population, and studies such as Wang et al. (2022) support that meaningful usability insights can still be achieved with small samples. The feedback provides a correct and context-specific evaluation of how well the system meets the operational needs of the Guidance Office. According to Suria (2024), the SUS is a statistically reliable tool in evaluating academic online systems, hence suitable for assessing satisfaction, ease of navigation, and operational efficiency.

Bachelor of Science in Business Administration continued to attract a substantial number of students with 102 enrollees, while 33 students chose Bachelor of Physical Education. In addition, 25 students enrolled in Bachelor of Elementary Education, and 77 students pursued Bachelor of Secondary Education. Lastly, 24 examinees were categorized under "No Course," representing students who did not enroll after taking the ICAT.

Table 3. Breakdown of ICAT examinees by course for Academic Year 2025–2026

Academic Program	Number of Students
Bachelor of Public Administration	47
Bachelor of Science in Information Technology	126
Bachelor of Science in Entrepreneurship	75
Bachelor of Arts in Social Science	126
Bachelor of Arts in Psychology	77
Bachelor of Arts in English Language	31
Bachelor of Science in Mathematics	8
Bachelor of Science in Business Administration	257
Bachelor of Physical Education	26
Bachelor of Elementary Education	27
Bachelor of Secondary Education	126
No Course (Did Not Enroll)	62
TOTAL	**1,001**

The usability of the proposed system was evaluated using the System Usability Scale (SUS), with the Guidance Counselor and Guidance Associate serving as the primary

Table 2. Breakdown of ICAT examinees by course for Academic Year 2024–2025

Academic Program	Number of Students
Bachelor of Public Administration	47
Bachelor of Science in Information Technology	126
Bachelor of Science in Entrepreneurship	75
Bachelor of Arts in Social Science	126
Bachelor of Arts in Psychology	77
Bachelor of Arts in English Language	31
Bachelor of Science in Mathematics	8
Bachelor of Science in Business Administration	257
Bachelor of Physical Education	26
Bachelor of Elementary Education	27
Bachelor of Secondary Education	126
No Course (Did Not Enroll)	62
TOTAL	**1,001**

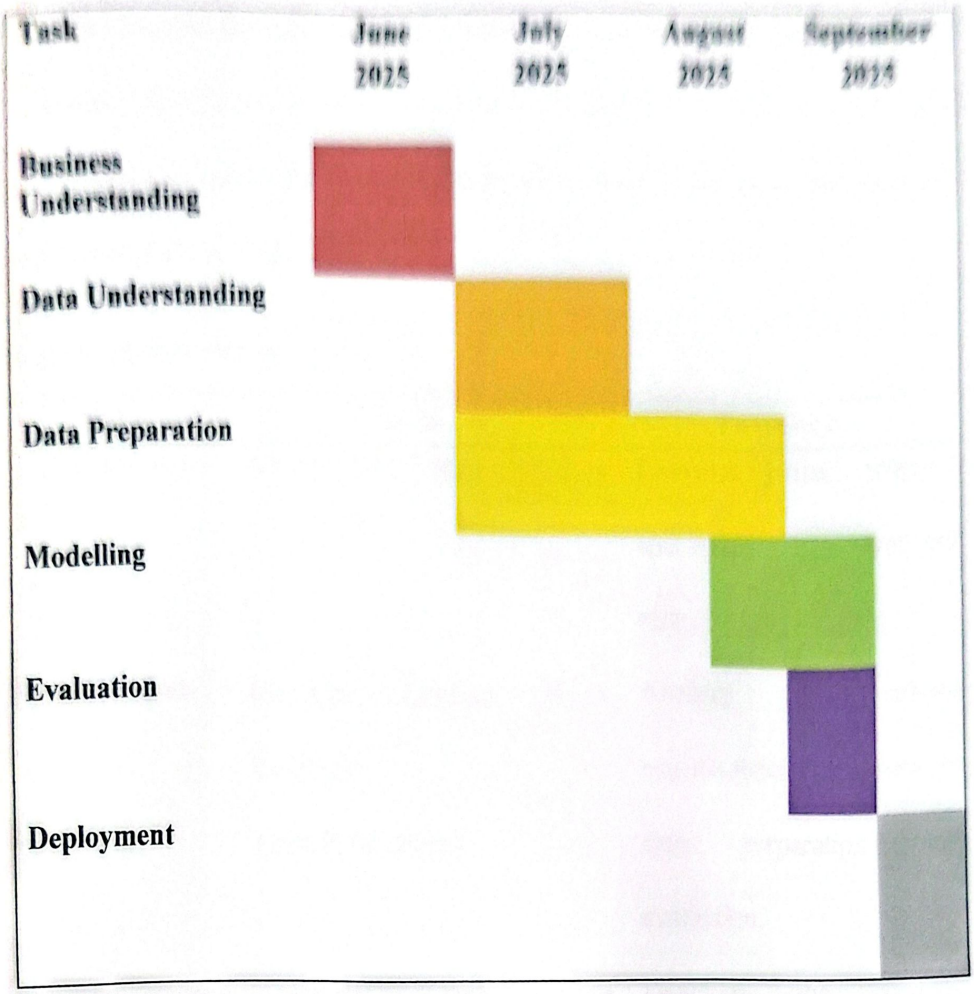
Table 3 shows the breakdown of ICAT examinees by course for Academic Year 2025–2026, based on a total of 510 student records. For this academic year, 14 students enrolled in the Bachelor of Public Administration, while 100 students pursued the Bachelor of Science in Information Technology. The Bachelor of Science in Entrepreneurship recorded 8 students, and 91 students enrolled in the Bachelor of Arts in Social Science. Moreover, 27 students pursued Bachelor of Arts in Psychology, and 9 students enrolled in Bachelor of Arts in English Language. Notably, no students enrolled in the Bachelor of Science in Mathematics during this academic year. The

Population and Locale of the Study

The study was conducted at Ilocos Sur Polytechnic State College (ISPSC), Tagudin Campus and focused on the implementation of a clustering system for analyzing Ilocos Sur Polytechnic State College Admission Test (ICAT) scores. The population comprised all students who took the ICAT during Academic Years 2024–2025 and 2025–2026. A total enumeration sampling technique was employed to include all ICAT examinees within the specified academic years.

Table 2 presents the detailed breakdown of ICAT examinees by course for Academic Year 2024–2025, with a total of 1,001 student records included in the analysis. During this academic year, 47 students were enrolled in the Bachelor of Public Administration, while 126 students pursued the Bachelor of Science in Information Technology. The Bachelor of Science in Entrepreneurship program recorded 75 enrollees, and another 126 students were enrolled in the Bachelor of Arts in Social Science. Additionally, 77 students took Bachelor of Arts in Psychology, whereas 31 students enrolled in Bachelor of Arts in English Language. The Bachelor of Science in Mathematics had the smallest enrollment with 8 students. In contrast, the Bachelor of Science in Business Administration recorded the highest number of enrollees at 257 students, indicating strong student preference for the program. Meanwhile, 26 students enrolled in Bachelor of Physical Education, 27 students in Bachelor of Elementary Education, and 126 students in Bachelor of Secondary Education. Furthermore, 62 examinees did not proceed with enrollment and were classified under "No Course," providing insight into non-enrollment behavior following the ICAT.

Project Schedule



Legend:

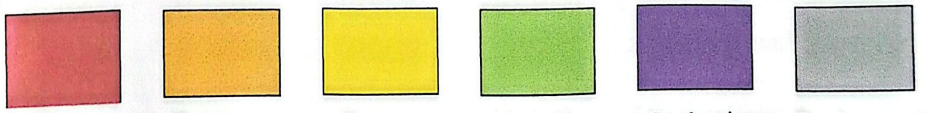


Figure 3. Project Schedule

Figure 3 illustrates the complete project schedule, showing the start and end dates of key activities, their interdependencies, and the overall progression of the project from inception to completion.

Chapter 3

METHODOLOGY

This chapter describes the methods and procedures utilized in the development and evaluation of the clustering system for ICAT scores. The research design, the development process, software model, project schedule, participants, research instruments, and data analysis techniques are discussed in this chapter to ensure that the study is done orderly and the results are valid, reliable, and appropriate for the research objectives.

Research Design

This study employed a Descriptive Developmental Research Design, which is appropriate for the development and evaluation of an educational system that integrates data analysis and system implementation. The descriptive component focused on examining archived ICAT results to identify patterns and trends in student performance without manipulating the original data. These findings served as the empirical basis for designing the clustering model and determining relevant performance indicators.

The developmental component involved the design, implementation, and refinement of a machine learning–based clustering system using Python and the Scikit-learn library's K-Means algorithm. Insights derived from the descriptive analysis informed how the algorithm grouped students based on similarities in their ICAT performance. Through this approach, descriptive analysis and machine learning complemented each other, with descriptive findings guiding model development and K-Means clustering generating meaningful student groupings.

The developed system presents its outputs through a web-based dashboard that allows authorized users to upload ICAT results, view cluster assignments, and analyze

strengths and areas that need improvement, thereby enhancing academic guidance and decision-making processes.

The study provides the **Guidance Office and faculty** of ISPSC Tagudin Campus with an efficient and reliable tool for interpreting ICAT results. The system helps identify students who may require academic assistance, recommend suitable academic tracks or courses, and support the screening of students for eligibility in board-related programs, leading to more informed and timely interventions.

The study also benefits **Teachers and Administrators** by generating organized student profiles and clustering results. These outputs allow the institution to deliver faster, more responsive, and evidence-based student services, reducing reliance on manual interpretation and improving the overall accuracy of academic evaluation.

The **Community** will benefit from this study as it promotes proper student placement based on academic merit and aptitude. This contributes to the development of well-prepared graduates who are more capable of contributing positively to national growth and workforce development.

The **Future Researchers** may use the results of this study as a foundation for further research in educational data mining, artificial intelligence, and machine learning applications in academic profiling. The system and methodology may be enhanced by integrating additional datasets or applying other clustering algorithms to further improve the analysis and understanding of student performance.

shall mainly discuss the development of automation of the analysis and visualization of ICAT data using machine learning techniques, focusing on efficiency, interpretability, and usability of guidance counselors and administrators. It aimed to serve as a decision-support tool complementary to existing assessment practices to provide empirical support for student profiling and academic management at the Tagudin Campus.

The study only utilizes the results of ICAT from the students of ISPSC Tagudin Campus. It does not include data from other campuses or data from other sources. It will also not consider aspects such as personality, family income level, and behavior. This study shall only focus on the grouping of students concerning their cognitive aptitude. The veracity of the system's clustering output is only as good as the completeness and validity of the ICAT data provided to the system. Since this is a web-based system, a stable internet connection is needed for the system to work at its best performance and for better access. In addition, the system is only designed for academic guidance and research purposes; thus, it does not supplant existing institutional evaluation or admission procedures. Possible future enhancements in the system include the integration of other datasets, the inclusion of other machine learning algorithms like DBSCAN or GMM, and its expansion to other campuses of ISPSC for wider validation and analysis.

Importance of the Study

Importance of the Study the ICAT Clustering System for ISPSC Tagudin Campus was developed to provide a data-driven approach in understanding student aptitude through the application of machine learning algorithms. This system supports the institution by offering a clearer, more objective assessment of students' cognitive

core cognitive areas such as General Ability, Verbal Aptitude, and Numerical Aptitude.

Specifically, this study aimed to:

1. examine the current process of interpreting and utilizing ICAT scores at ISPSC Tagudin Campus;
2. design and develop a clustering model using the K-Means algorithm to categorize students based on their ICAT results and define performance-based groupings;
3. measure the accuracy and performance of the K-Means clustering algorithm using cluster validation metrics such as the Silhouette Score, Calinski–Harabasz Index, and Davies–Bouldin Index;
4. develop a web-based system for the Guidance Office that stores ICAT scores, recommends suitable academic programs based on student profiles, and provides a dashboard that visualizes clustering results; and,
5. evaluate the usability and effectiveness of the developed system in generating meaningful and accurate student profiles for academic guidance.

Scope and Limitation of the Study

This study is concerned with the design, development, and implementation of a web-based system that clusters students of Ilocos Sur Polytechnic State College (ISPSC) Tagudin Campus based on their ICAT percentile scores. The proposed system uses the K-Means Clustering Algorithm to analyze the results of the students in six cognitive areas, namely: General Ability, Verbal Aptitude, Numerical Aptitude, Spatial Aptitude, Perceptual Aptitude, and Manual Dexterity. From this clustering process, the system will show a pattern of strengths and weaknesses among the students that could be useful in academic advising, course recommendation, and institutional planning. The study

The Process Phase was mainly concerned with the processing of raw ICAT data into useful results by applying the K-Means Clustering Algorithm under the system named Clustering Student ICAT Score Using Machine Learning Algorithm. In this phase, the gathered ICAT data were processed for similarities and patterns existing in scores among the students. The algorithm clustered the students based on the dominant aptitude pattern in all six ICAT areas so that the classification of a student to a cluster could define his strengths and learning tendencies.

The Output Phase showed the result from the clustering process. In this case, it included the created categorized student clusters based on the dominant aptitude patterns and individual student profiles that highlighted strengths and risks. These were made possible by an administrative dashboard that was shown to enable viewing and interpretation of clustering outcomes. Using the system, the students qualified to enter board programs were identified, academic tracks or courses should be recommended, and a data-driven decision support system should be established to help the Guidance Office and academic units in academic planning and decision making.

Objectives of the Study

The main objective of this study, titled "Clustering Student ICAT Scores Using Machine Learning Algorithm," was to develop and implement a web-based system that analyzed and grouped the ICAT scores of students at Ilocos Sur Polytechnic State College (ISPSC) Tagudin Campus using machine learning techniques. The system was designed to identify and categorize students according to their dominant cognitive abilities and learning profiles, providing a data-driven basis for academic guidance and course recommendations. Furthermore, the study sought to assist in determining student eligibility for board courses by identifying those who fell below critical thresholds in

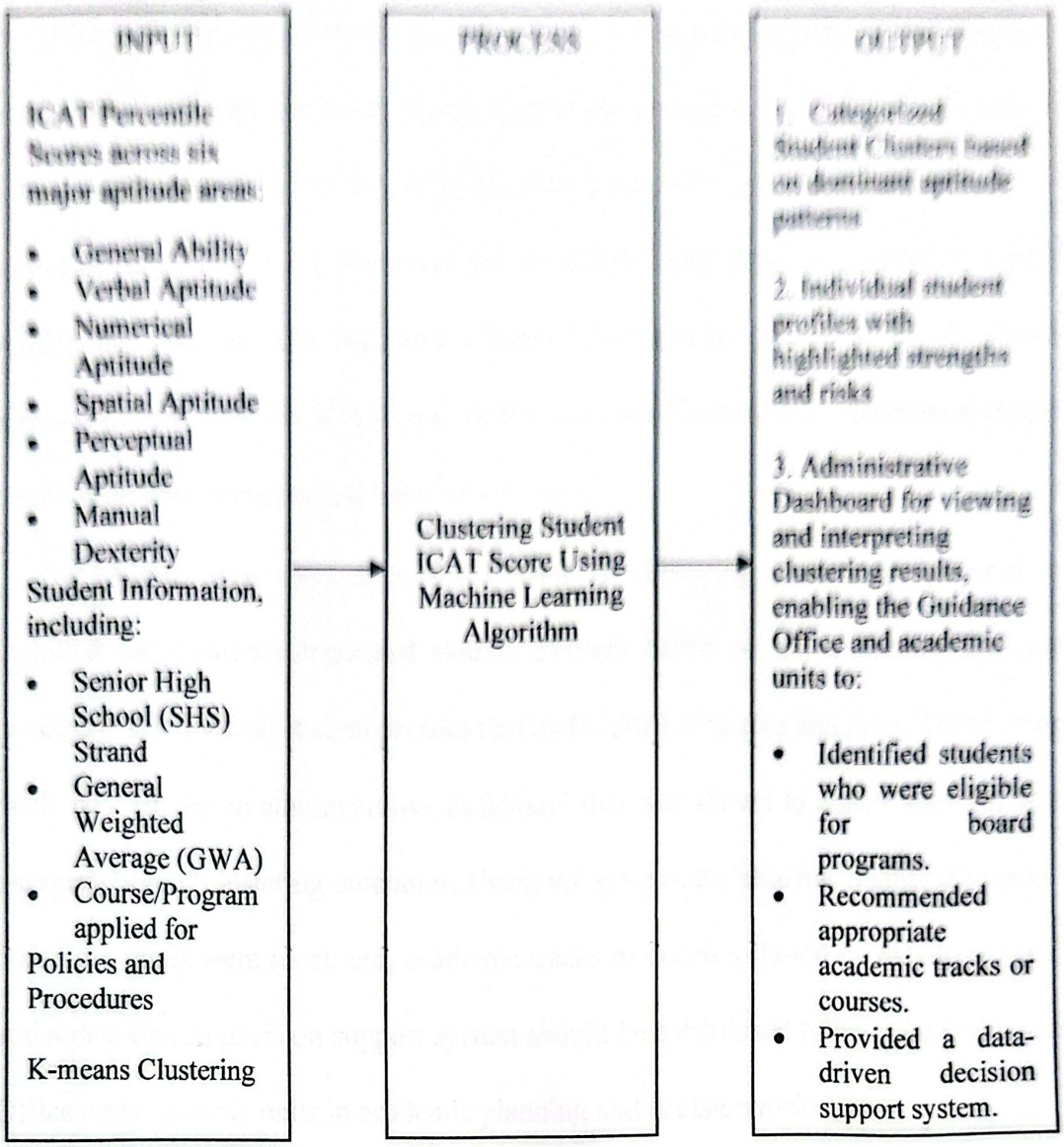


Figure 1. Conceptual Framework

As shown in figure 1, The Input Phase of the study involved the ICAT percentile scores over six big aptitude areas, namely: General Ability, Verbal Aptitude, Numerical Aptitude, Spatial Aptitude, Perceptual Aptitude, and Manual Dexterity. In addition to these test results, student information regarding SHS strand, GWA, and course or program applied to, along with existing policies and procedures, were also prepared. All of the data were gathered from the ISPSC Guidance Office for accuracy and confidentiality. All these served as the input basis for the analysis of aptitude and academic background of students in a data-driven manner.

segregation of students into groups with profiles of similarities in cognition and aptitude. The algorithm systematically identifies these clusters reflecting each distinct strength, weakness, and learning tendency. The output of the proposed framework encompasses categorical student clusters that underscore differing cognitive profiles along with potential academic risks. These outputs are designed to support the ISPSC Tagudin Campus in devising targeted academic interventions, improved guidance services, and the adoption of advising strategies that better mirror the students' learning potential. Furthermore, the framework encourages data-informed decision-making by ensuring that administrators and faculty stakeholders are provided with unbiased analysis outcomes. It also allows for the timely detection of students who need extra learning support or special attention. Overall, the framework developed using IPO ensures a systematic and sustainable methodology for institutional action that stems from assessment data.

Dexterity and Spatial Aptitude. Other emerging groups may include Language-Dominant or Math-Dominant learners, characterized by higher performance in verbal or numerical reasoning, as well as students with Visuospatial Strengths or Mixed Profiles, where notable strengths in certain areas coexist with lower performance in others. This clustering approach allows for a more holistic student profiling process by identifying not only areas of strength but also learning gaps that may require academic support or guidance. By using K-Means clustering as the core analytical method, the study enables the automatic grouping of students based on similarity in aptitude patterns without relying on predefined academic categories. ICAT scores serve as the primary input data, allowing the system to discover natural groupings that reflect actual student performance characteristics. As a result, this approach supports objective, data-driven, and scalable student profiling that can enhance academic advising, course alignment, and institutional decision-making.

Ultimately, it contributes to a smarter and more adaptive education system, one that aligns the strengths of students with the appropriate learning paths and even future career trajectories.

Conceptual Framework of the Study

This study adopts the Input-Process-Output Model as a conceptual framework, providing an organized structure of how the data of students are transformed into actionable academic insights. The input would be the ICAT percentile scores and the corresponding student background information gathered from the ISPSC Guidance Office, which serves as the foundational dataset to be analyzed. The process involves machine learning techniques, particularly the use of the K-Means Clustering Algorithm to analyze the results of the ICAT in order to uncover any hidden patterns and for the

Campus. While the ICAT is systematically administered and interpreted for admission and program recommendation purposes, the resulting data remains largely underexplored beyond these immediate functions. With the increasing availability of educational data and the growing demand for data-driven decision-making in higher education, there is a need to transform traditional aptitude test results into actionable insights. This study recognizes that manual interpretation alone limits the potential of ICAT data to reveal deeper patterns of students' cognitive and psychomotor strengths. By applying machine learning specifically K-Means clustering this research aimed to automate student profiling, uncover latent groupings within ICAT scores, and provide a more holistic understanding of student aptitude. Ultimately, the capstone project was undertaken to enhance academic guidance, support evidence-based curriculum planning, and contribute to a more adaptive and personalized educational system aligned with students' inherent capabilities and learning potential.

Addressing this shortfall, this research proposes a machine learning–based clustering approach to analyze and group students according to their ICAT scores. Rather than focusing solely on overall test results, the proposed method examines patterns of strengths and weaknesses across the six ICAT aptitude areas to provide a deeper and more meaningful interpretation of student abilities. Through this approach, students can be organized into distinct groups that reflect their cognitive and practical tendencies. For example, All-Around Performers demonstrate consistently balanced performance across all aptitude areas, while Cognitive-Oriented Learners show stronger capabilities in analytical and reasoning-based domains such as General Ability, Verbal Aptitude, and Numerical Aptitude. In contrast, Practical-Oriented Learners tend to perform better in hands-on and applied domains, particularly Manual

student, enabling the identification of their strengths, weaknesses, and learning tendencies.

The ICAT is a standardized test that ISPSC purchases from accredited psychological centers specializing in aptitude testing. These centers make sure that the test is valid, reliable, and follows professional standards. It is administered by the Guidance Office of ISPSC Tagudin Campus and keeps the records of all its examinees. Immediately after the examination, personal data and raw scores of students are manually checked and converted into percentile ranks using the official ICAT manual containing established scoring norms. These norms enable counselors to interpret the results accurately as informed guidance for the students. Usually, those who obtained scores at or above the 33rd percentile in the core cognitive areas such as General Ability, Verbal Aptitude, and Numerical Aptitude qualify in the board courses. Furthermore, the student's Senior High School General Weighted Average is considered in recommending them to suitable academic programs. Practically, after taking the ICAT, the students' test results are encoded into a digital template after their personal data and raw scores have been manually verified. These are then printed and given, one copy for the student and one copy each for the academic department concerned and one for the official record in the Guidance Office. In this procedure, although accurate documentation is ensured, the results were then underutilized; it tends to be used only for admission purposes and not for deeper analyses that may reveal meaningful insights into the potentials for learning among students.

This capstone project was conducted to address the existing gap between the collection of ICAT results and their meaningful utilization for student development and institutional decision-making at Ilocos Sur Polytechnic State College–Tagudin

At the Ilocos Sur Polytechnic State College Tagudin Campus, one of the essential tools used in knowing the aptitude of the students is the ISPSC College Admission Test or the ICAT. The test measures students' capability in six major areas: General Ability, Verbal Aptitude, Numerical Aptitude, Manual Dexterity, Spatial Aptitude, and Perceptual Aptitude. Unlike traditional tests that determine what students have learned already, the ICAT examines their potentials to learn and to perform a particular task given proper teaching and training. As explained by MSEd (2024), "an aptitude test is designed to assess what a person is capable of doing or to predict what a person is able to learn or do give the right education and instruction." Hence, the purpose of the ICAT is not to label the students but to serve as a guide for teachers about the students' potentials for learning and development. According to the ISPSC Guidance Office, the ICAT has several subtests, each with a distinct area of ability that it seeks to measure. General Ability measures logical thinking and problem-solving skills. It offers a general overview of a student's intellectual potential. Verbal Aptitude will determine understanding of words, reading comprehension, and communication skills; all are vital in language and communication-related courses. Numerical Aptitude tests the ability to reason with numbers and to solve mathematical problems, which are essential reasoning skills in most analytical and quantitative courses. Spatial Aptitude measures the capacity to visualize and manipulate shapes and patterns, an ability important in architecture, design, and engineering studies. In the field of technical and data-related fields, Perceptual Aptitude measures alertness, accuracy, and attention to detail. Lastly, Manual Dexterity measures coordination and fine motor control, reflecting one's capability to perform practical and hands-on tasks efficiently. In sum, these subtests generate a complete profile of the cognitive and psychomotor capabilities of each

2025). By clustering students according to their dominant ICAT dimensions, this study is grounded in Gardner's principle that learners do better when their education matches their natural cognitive strengths.

At the national level, the increasing availability of student-related data in higher education institutions has encouraged the adoption of data-driven approaches to better understand learner performance and competencies. Several studies have demonstrated the effectiveness of clustering techniques in analyzing educational datasets and supporting academic decision-making. Lee and Lee (2024) examined various clustering algorithms for segmenting college students based on competency indicators and found K-Means clustering to be the most stable and reliable approach for grouping students with similar characteristics. Similarly, Golovnev and Kaneva (2023) illustrated how clustering methods can be used to systematically organize academic disciplines into meaningful thematic groups, highlighting the flexibility and applicability of such techniques in handling large and complex educational data. In response to recognized limitations of traditional K-Means clustering, Wan et al. (2022) proposed an enhanced version of the algorithm that improved analytical accuracy in student-related data analysis. These studies collectively emphasize the growing role of machine learning and clustering methods in educational research and institutional planning. Drawing from these national and international research efforts, the present study applies K-Means clustering to the analysis of ICAT results in order to automate and enhance student profiling. By utilizing a data-driven framework, the study seeks to contribute to more personalized academic advising, informed curriculum development, and evidence-based decision-making within higher education institutions in the Philippines.

Chapter 1

INTRODUCTION

Background of the Study

The present scenario of higher education is faced with so much pressure to make decisions about students' aptitudes and cognitive profiles effectively for educational planning and career guidance. Various related studies have established the potential of clustering in educational data analysis. Fathanah (2022) showed that there was an enhancement in the performance of students' writing after being exposed to clustering techniques, which helped them to organize ideas more coherently, hence its potential to support learning. Similarly, Nafuri et al. (2022) applied K-Means clustering to group B40 students in Malaysia based on academic outcomes, identifying patterns in dropout risk and graduation rate. Their study showed that optimized clustering models may present performance trends informative in educational planning. Agha, Meghji, and Bhatti (2023) also applied K-Means to analyze the grades of students at Mehran University of Engineering and Technology, identifying distinct academic patterns across different subjects. Their findings highlighted that clustering was key in finding the hidden relationships within student performance data. Kaban et al. (2023) further compared several clustering algorithms and found that on normalized datasets of educational data, K-Means performed well, though algorithm performances depend on dataset complexity. This stresses the importance of choosing the right techniques to get accurate results. The theoretical underpinning of this approach to clustering is influenced by Howard Gardner's Theory of Multiple Intelligences, which states that individuals possess various types of intellectual capabilities, such as logical-mathematical, verbal-linguistic, bodily-kinesthetic, and spatial intelligence (MSEd,

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DEDICATION

This work is a testament to my growth as an Information Technology student and a reflection of my continuous journey toward becoming a professional in the IT industry. It symbolizes the lessons, challenges, and experiences that have shaped me both academically and personally.

To my parents, thank you for your unwavering support, constant belief in my abilities, sacrifices, and endless encouragement. You have been my greatest source of strength and motivation throughout this journey.

To the Atelo family, I am deeply grateful for the friendship, inspiration, and for being one of the reasons I strive to grow and improve myself each day.

Lastly, to Almighty God, for the wisdom, guidance, and blessings that made all of this possible this achievement is dedicated to you.

Azriel Jon M. Yukee

ACKNOWLEDGEMENT

The researchers sincerely thank Almighty God for granting them wisdom, guidance, and strength to accomplish this study. His presence has been their source of perseverance and inspiration.

They extend their heartfelt gratitude to Dr. George O. Villanueva, DIT, their adviser, for his continuous support, guidance, and valuable insights that contributed greatly to this research.

Appreciation is also given to the Guidance Office Staff of Ilocos Sur Polytechnic State College – Tagudin Campus for their cooperation and valuable feedback through the Google Form survey, which was essential in evaluating the system's usability and effectiveness.

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Lastly, the researchers thank their parents and families for their unwavering love, support, and understanding, which served as their foundation in completing this study.

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A.P.M

ABSTRACT

YUKEE, AZRIEL JON M., BONIFACIO, CHRISTINE LORENCE., SALVADOR, JOHN MARK A., MACABITAS, ARIES P., A "CLUSTERING STUDENT ICAT SCORE USING MACHINE LEARNING ALGORITHM," Ilocos Sur Polytechnic State College, Tagudin, Ilocos Sur, November 3, 2025.

Adviser: **GEORGE VILLANUEVA, DIT**

The study developed and evaluated a web-based system that utilized a machine learning clustering algorithm applied to the students' ICAT scores in order to find patterns in academic performance. It was designed to help an institution, through the Guidance office and faculty, to classify students based on competency levels and understand their academic strength and weakness. Following the framework of the CRISP-DM, analysis of existing ICAT score interpretation practices, clustering system development, applying the K-Means algorithm, and assessment of system usability were conducted. The sum result of the archived ICAT data at Ilocos Sur Polytechnic State College–Tagudin Campus was used for the analysis. Analysis on the usability test using the System Usability Scale gave a mean score of 68.75 with good ratings on effectiveness, efficiency, and user satisfaction. The K-Means algorithm showed successful determination of optimal clusters into four based on aptitude components, the number of clusters confirmed by the Elbow method, and cluster quality on valuation metrics with a strong meaning. In general, the findings confirm that the system developed in this study is effective in generating meaningful groupings about student performance, supportive of data-driven academic profiling, guidance, placement, and institutional decision-making.

Keywords: Clustering, Machine Learning Algorithm, Ilocos Sur Polytechnic State College College Admission Test (ICAT), CRISP-DM, System Usability Scale (SUS)

APPROVAL SHEET

This Capstone Project entitled CLUSTERING STUDENT ICAT SCORE USING MACHINE LEARNING ALGORITHM was prepared and submitted by **AZRIEL JON M. YUKEE, CHRISTINE LORENCE H. BONIFACIO, JOHN MARK A. SALVADOR, ARIES P. MACABITAS** in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, and has been examined and recommended for acceptance by the Oral Examination Committee (OREC) on November 3, 2025, composed of:

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October 20, 2025

DR. OMAR C. VALENZUELA
Guidance Counselor

Sir

The undersigned students of the Bachelor of Science in Information Technology (BSIT) program respectfully seek your approval for the deployment of our Capstone Project 2 study entitled "Clustering Student ICAT Score Using Machine Learning Algorithm," to be implemented in the Guidance Office of Ilocos Sur Polytechnic State College (ISPSC) – Tagudin Campus.

The deployment process will include system installation, configuration of the web application, creation of administrative accounts, user orientation, and pilot testing involving selected Guidance Office personnel and faculty members. All data to be processed will be handled with strict confidentiality, following institutional data privacy policies and ethical standards.

We sincerely appreciate your support and approval in deploying our system, which we believe will contribute significantly to ISPSC's ongoing efforts to integrate data-driven solutions in education and improve academic decision-making processes.

Respectfully yours,

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